

BPS spectra of $\text{Tr}[\Psi^p]$ matrix models for odd p

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Abstract

We study the BPS cohomology and Hamiltonian of a model built from an $N \times N$ matrix Ψ of complex fermions, with supercharge $Q_p = \text{Tr}(\Psi^p)$ for odd $p \geq 3$. Numerical calculations give the BPS multiplicity at every fermion number for $(p, N) = (5, 3), (5, 4), (5, 5), (7, 4)$, and through fermion number six at $(7, 5)$. In $Z_{\text{BPS}}^{(p, N)}(x) = \sum_R h_R x^R$, x records fermion number and h_R counts BPS states in sector R . In each of the four cases computed at every fermion number, factoring out the lowest power of x leaves a polynomial divisible by a positive power of p and by $(1+x)^N$. The quotient by these factors has nonnegative integer coefficients. Exact decoupling of $\text{Tr}\Psi$ and the Euler characteristic prove divisibility by $(1+x)^2$. Pairing sectors of complementary fermion number gives a third factor when N is odd. For odd $p \leq 2N - 1$, the only range in which Q_p can be nonzero, the same pairing proves the corresponding rank symmetry for Q_p . The additional factors required to reach $(1+x)^N$ remain conjectural in general. Conditional on reported numerical vanishings at larger N , the projections identify as fortuitous those BPS classes with no preimage from some larger matrix size. Evaluating at $x = 1$ gives the total BPS count $Z_{\text{BPS}}^{(p, N)}(1)$. For each fixed odd p , the Witten indices imply that, for any sequence of matrix sizes $N_j \rightarrow \infty$ along which $N_j^{-2} \log Z_{\text{BPS}}^{(p, N_j)}(1)$ converges, its limit lies in $[\log(2 \cos \frac{\pi}{2p}), \log 2]$. After dividing the Hamiltonian by p^2 , normal ordering gives an alternating sum of operators K_k , with K_k containing k creation and k annihilation operators. We obtain exact formulas for K_0, K_1, K_2 when $p = 5$ and $N \geq 3$. Only K_3, K_4 can contain spectral information beyond the number of traceless fermions and the quadratic Casimir. At $N = 3$, the Hodge star maps the invariant cubic form to the quintic form up to scale, so all five terms commute. Calculations with integer matrices give nonzero commutators in the reported $N = 4$ sectors.

1 Introduction

We study the quantum mechanics of a single $N \times N$ matrix Ψ whose entries Ψ_{ij} are fermion creation operators. The group $\text{SU}(N)$ is a global symmetry and is not gauged, so the full Hilbert space is retained. The supercharge and Hamiltonian are

$$Q_p = \text{Tr}(\Psi^p), \quad p \geq 3 \text{ odd}, \quad (1)$$

where $\{\Psi_{ij}, \Psi_{kl}^\dagger\} = \delta_{ik}\delta_{jl}$ and $H = \{Q_p, Q_p^\dagger\}$ [1, 2]. The Hilbert space is the fermionic Fock space $\mathcal{H} = \Lambda^\bullet(\mathbb{C}^{N \times N})$, whose component of degree R has fermion number R . For odd p , $Q_p^2 = 0$. Since $H = \{Q_p, Q_p^\dagger\}$, the BPS space is $\ker Q_p / \text{im } Q_p$: each cohomology class has a representative of zero energy, and every state of zero energy defines such a class.

We determine the BPS multiplicities at finite N , compare BPS classes under projections between ranks, and identify the terms that can contain spectral information not fixed by the quadratic Casimir.

For $p = 3$, Chen obtained the exact total generating function [3]. In this case the Hamiltonian is a constant minus the quadratic Casimir, in accordance with Kostant's analysis of the Laplacian

and Casimir spectrum on an exterior algebra [4]. Reference [3] also noted the extensions $Q_p = \text{Tr}(\Psi^p)$ for odd p , but did not study $p \geq 5$ in detail. For $p \geq 5$, the Hamiltonian is not in general determined by the central Casimirs. The numerical example at $(p, N) = (5, 4)$ gives two equivalent $\text{SU}(4)$ representations at different energies.

After normal ordering, the reduced Hamiltonian is $\tilde{H} := H/p^2 = \sum_{k=0}^{p-1} (-1)^k K_k$, where K_k contains k creation and k annihilation operators. For $p = 5$, we determine K_0, K_1, K_2 for every $N \geq 3$. Their alternating sum depends only on the number of traceless fermions and the quadratic Casimir. Only K_3 and K_4 can contribute additional spectral information not fixed by these operators. In the numerical $(5, 4)$, $R = 4$ example, the splitting of the two equivalent 20-dimensional blocks in $\tilde{H}$ is produced by $-K_3 + K_4$. At $N = 3$, the traceless space for one particle has dimension eight. The Hodge star on its exterior algebra sends the invariant cubic form to a nonzero multiple of the invariant quintic form, making all five K_k commute. Calculations with integer matrices show that they do not all commute at $N = 4$, but this does not decide whether the model is chaotic or integrable.

The present model is not intended as a holographic theory. It provides a finite dimensional system in which the BPS cohomology can be computed and compared as N changes. Within the cohomological analysis, we keep two diagnostics separate. The first is the difference between the BPS count and the lower bound from the Witten index in each class of fermion numbers modulo p . The second is whether a class belongs to the stable image of the projections from larger N . Neither property implies the other.

To formulate the second diagnostic, we adapt the distinction introduced in Ref. [5]. Here a class is monotone if it lies in the stable image, the intersection of the images on cohomology induced by projections from all larger matrices. A class outside it is fortuitous. That reference uses a covering complex, and we do not assume the two constructions are equivalent. This distinction has been studied in $\mathcal{N} = 4$ supersymmetric Yang–Mills theory and in supersymmetric Sachdev–Ye–Kitaev models [5–7]. Related constructions appear in the D1–D5 conformal field theory [8, 9] and in Aharony–Bergman–Jafferis–Maldacena theory [10, 11].

Further examples of fortuity have been studied in a model with higher spin symmetry [12] and in a supergravity analysis of the D1–D5 system [13]. A relevant deformation of $\mathcal{N} = 4$ Yang–Mills theory can produce fortuitous cohomology [14]. Quantum corrections can also pair and lift additional monotone and fortuitous classes [15].

At finite N , trace relations in coupled matrix and vector systems suggest a bosonic analogue of fortuity [16]. Secondary invariants in invariant rings at finite N have also been studied as algebraic models of nonperturbative states [17]. At $N = 3$, fortuitous black hole states at rank $N - 1$ were combined with dual giant gravitons to construct BPS cohomology at rank N [18]. At tree level, the cohomology of Q can differ between gauge theories related by S duality, with the reported mismatch involving one fortuitous class and one monotone class [19].

The calculations closest in method construct the supercharge complex separately at each charge and determine its cohomology for small values of N [5–8, 10]. A related comparison for the dynamics is the model of Ref. [20], in which a fixed supercharge replaces random couplings. Its agreement with supersymmetric Sachdev–Ye–Kitaev theory concerns the melonic approximation for $\text{SU}(2)$ singlet observables at small charge. The model studied here has an exact $\text{SU}(N)$ symmetry that acts by conjugation on the matrix fermions. No melonic limit is constructed here.

Other fermionic matrix models are discussed in Refs. [21–28], including the single matrix model of Ref. [29]. A different family of supersymmetric matrix models, motivated by the BMN model and by holography in plane wave backgrounds, was proposed in Ref. [30]. The invariant antisymmetric tensors used in the supercharges come from Lie algebra cohomology [31, 32].

We report numerical BPS multiplicities in every fermion number sector for $(p, N) = (5, 3)$, $(5, 4)$, $(5, 5)$, and $(7, 4)$. At $(7, 5)$, only the sectors $R \leq 6$ are computed. Section 4 explains the numerical rank assignments and their modular checks. Unless an analytic proof is given, the

resulting multiplicities are numerical rather than certified ranks over $\mathbf{Q}$.

Let h_R be the BPS multiplicity at fermion number R , and define $Z_{\text{BPS}}^{(p,N)}(x) = \sum_R h_R x^R$, with x recording fermion number. In each of the four cases computed at every fermion number, the numerical polynomial has the form

$$Z_{\text{BPS}}^{(p,N)}(x) = p^{b_{p,N}} x^{q_{\min}} (1+x)^N T_N^{(p)}(x).$$

Here $q_{\min}$ is the smallest fermion number carrying BPS states, $b_{p,N}$ is the exponent of the common positive power $p^{b_{p,N}}$, and $T_N^{(p)}$ has nonnegative coefficients. Exact trace mode decoupling and the Euler characteristic give at least two factors of $1+x$. Pairing complementary sectors in the highest exterior degree gives a third when N is odd. The factors still needed to reach $(1+x)^N$ are conjectural.

Write $r_R = \text{rank } Q_R$, where Q_R maps the sector with fermion number R to that with fermion number $R+p$. For odd $p \leq 2N-1$, the same pairing gives $r_R = r_{N^2-p-R}$. With $\mathcal{R}_{p,N}(x) = \sum_R r_R x^R$, the exact relation $(1+x) \mid \mathcal{R}_{5,5}(x)$ then determines the middle map rank at $(5,5)$ from the ten reported ranks below it.

The projections identify all BPS classes at $(5,3)$, and the classes in the sectors listed in Table 2 at $(5,4)$ and $(7,4)$, as fortuitous. These conclusions are conditional on the reported numerical vanishings at larger N .

Independently, set $x = 1$, so that $Z_{\text{BPS}}^{(p,N)}(1)$ is the total BPS count. Fix an odd p and let $N_j \rightarrow \infty$ be any sequence of matrix sizes for which $N_j^{-2} \log Z_{\text{BPS}}^{(p,N_j)}(1)$ converges. The Witten indices imply that the resulting limit lies in

$$\left[\log \left(2 \cos \frac{\pi}{2p} \right), \log 2 \right].$$

Thus $x = 1$ and p are held fixed. Only the matrix size tends to infinity.

Figure 1 shows the finite pairs (N, p) treated numerically in this paper.

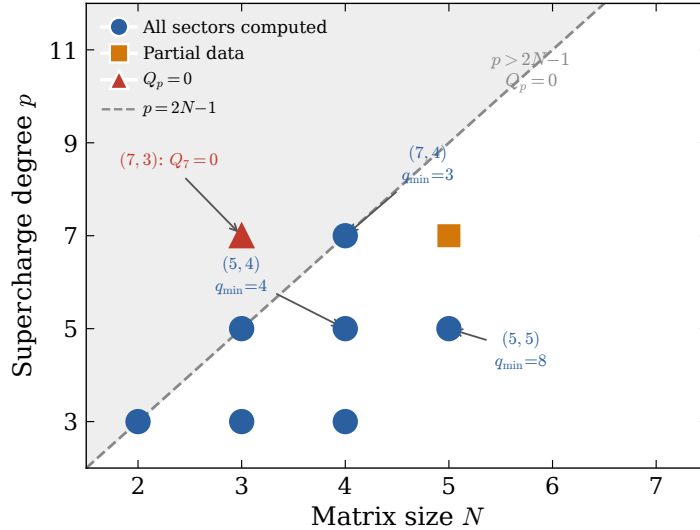


Figure 1: Cases considered in this paper. In the shaded region $p > 2N - 1$, one has $Q_p \equiv 0$. Blue circles mark cases with numerical BPS data in every sector. The orange square marks the partial data at $(7, 5)$, and the red triangle marks $(7, 3)$, where $Q_7 = 0$. Annotations show the minimum BPS charge $q_{\min}$ where known.

Section 2 defines the model and its cohomological structure. Section 3 derives the Hamiltonian after normal ordering and compares the commutators at $N = 3$ and $N = 4$. Section 4

describes the numerical method, presents the BPS spectra, and compares different values of N by projection. Section 5 states the conjecture and proves the exact identities for the ranks. Section 6 compares the results with supersymmetric Sachdev–Ye–Kitaev models, derives the bound at large N , and gives the conclusions and open questions. The appendices contain deferred proofs, algebraic background, and the sector data.

2 Model and cohomological structure

We describe the decomposition of the Fock space, the action of $SU(N)$, the cohomological description, and the exact decoupling of the trace fermion. The Hamiltonian is studied separately in Section 3.

2.1 Fock space and sector decomposition

The Hilbert space $\mathcal{H} = \Lambda^\bullet(\mathbb{C}^{N \times N})$ is the 2^{N^2} -dimensional Fock space built from the N^2 complex fermion modes Ψ_{ij} . We use the convention that Ψ_{ij} acts by exterior multiplication and $\Psi_{ij}^\dagger$ by contraction, so

$$\Psi_{ij}^\dagger |0\rangle = 0, \quad F := \sum_{i,j} \Psi_{ij} \Psi_{ij}^\dagger,$$

where $|0\rangle$ is the vacuum and F counts fermion number. We separate the trace and traceless creation modes by

$$\chi := \frac{1}{\sqrt{N}} \text{Tr} \Psi, \quad A := \Psi - \frac{\chi}{\sqrt{N}} \mathbf{1}, \quad \text{Tr} A = 0. \quad (2)$$

They obey

$$\{\chi, \chi^\dagger\} = 1, \quad \{\chi, A_{ij}^\dagger\} = 0, \quad \{A_{ij}, A_{kl}^\dagger\} = \delta_{ik} \delta_{jl} - \frac{1}{N} \delta_{ij} \delta_{kl}. \quad (3)$$

Thus the traceless space for one particle is $W_0 = \{X \in \text{Mat}_N(\mathbb{C}) : \text{Tr} X = 0\} \simeq \mathfrak{sl}_N(\mathbb{C})$, with $SU(N)$ acting by conjugation. Throughout, $\text{Mat}_N(\mathbb{C})$ carries the Hilbert–Schmidt inner product $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$, for which the matrix units E_{ij} are orthonormal. Here and below, “orthonormal” refers to this inner product. Choose an orthonormal Hermitian basis $\{T_\alpha\}_{\alpha=1}^{N^2-1}$ of W_0 and write

$$A = \sum_{\alpha=1}^{N^2-1} b_\alpha^\dagger T_\alpha, \quad b_\alpha = (b_\alpha^\dagger)^\dagger.$$

The operators $b_\alpha^\dagger$ are the independent traceless fermion creation modes. They satisfy $\{b_\alpha, b_\beta^\dagger\} = \delta_{\alpha\beta}$. All mixed anticommutators between the trace and traceless modes vanish. In particular, $\{\chi, b_\alpha\} = \{\chi^\dagger, b_\alpha^\dagger\} = 0$. With $T_0 = \mathbf{1}/\sqrt{N}$, the space for one particle therefore decomposes orthogonally as $\text{Mat}_N(\mathbb{C}) = \mathbb{C} T_0 \oplus W_0$, and χ is the creation operator associated with T_0 . In this convention the occupation number of the trace mode is $n_\chi := \chi \chi^\dagger$, and $\hat{n} := F - n_\chi$ counts the traceless fermions. With $V_R := \Lambda^R(\mathbb{C}^{N \times N})$ the sector with fermion number R , the Hilbert space decomposes as

$$\mathcal{H} = \bigoplus_{R=0}^{N^2} V_R, \quad \dim V_R = \binom{N^2}{R}.$$

By construction, $F|_{V_R} = R$. Since Q_p raises fermion number by p , its sectorwise action is a map

$$Q_R : V_R \rightarrow V_{R+p}. \quad (4)$$

We use the convention $V_R = 0$ outside $0 \leq R \leq N^2$, so $Q_R = 0$ whenever its source or target sector lies outside this range. The Hamiltonian restricted to V_R is

$$H_R = Q_R^\dagger Q_R + Q_{R-p} Q_{R-p}^\dagger. \quad (5)$$

Particle–hole conjugation is the unitary Hodge map on the exterior algebra, sending each occupation pattern to its complement. Its phase is chosen to exchange creation and annihilation operators. It sends V_R to V_{N^2-R} and conjugates Q_p to $Q_p^\dagger$ up to an irrelevant phase. Consequently it maps H_R to H_{N^2-R} , giving

$$h_R = h_{N^2-R} \quad (6)$$

for the multiplicities h_R at zero energy. The BPS generating function graded by fermion number is

$$Z_{\text{BPS}}(x) := \sum_{R=0}^{N^2} h_R x^R, \quad (7)$$

where x is the fugacity for fermion number. We write

$$Z_{\text{BPS}} := Z_{\text{BPS}}(1) = \sum_{R=0}^{N^2} h_R \quad (8)$$

for the total BPS count, and use the unevaluated symbol $Z_{\text{BPS}}(x)$ whenever the charge grading matters. The minimum BPS charge $q_{\min}$ is the smallest R with $h_R > 0$.

2.2 Quadratic Casimir and the spectrum

For $p = 3$, the supercharge Q_3 on the traceless sector $\Lambda^\bullet W_0 \simeq \Lambda^\bullet(\mathfrak{su}(N) \otimes_{\mathbb{R}} \mathbb{C})$ is wedge multiplication by the cubic primitive cocycle. Here a primitive cocycle is an indecomposable generator of the cohomology of the Lie algebra. In this model, it is the invariant antisymmetric tensor whose components are the coefficients of $\text{Tr}(A^3)$. In the normalization used here, Chen’s result is

$$H = 3N(N^2 - 1) - 9\hat{C}_2. \quad (9)$$

Here $\hat{C}_2$ is the quadratic Casimir for the conjugation action on the full Fock space. The trace mode is an $\text{SU}(N)$ singlet, and $\hat{C}_2$ is normalized to have eigenvalue N on the adjoint representation. Kostant studied the corresponding Casimir and Laplacian spectrum on the exterior algebra of a compact Lie algebra [4]. Every state in a fixed irreducible summand therefore has the same energy. Representation theory can consequently determine the cubic spectrum, as in Ref. [3].

For odd $p \geq 5$ with $p \leq 2N - 1$, the trace fermion still decouples exactly (Section 2.4), and the traceless supercharge is defined by the primitive cocycle of degree p [31, 32]. The associated Laplacian is not in general a function only of the central Casimir operators [33, 34]. The numerical example in Section 4.2 makes this failure explicit. Two equivalent irreducible representations occur at different energies.

Cartan generators. Adding the Cartan generators does not resolve repeated irreducible summands. They commute with H , since H is invariant under $\text{SU}(N)$, and label the weight of a state inside an irreducible representation. A repeated representation, however, appears as $\mathbb{C}^m \otimes V_\lambda$ with $m \geq 2$, and every element of the Cartan subalgebra acts as $\mathbf{1}_{\mathbb{C}^m} \otimes \rho_\lambda(X)$ for X in the Cartan subalgebra. It acts trivially on the multiplicity space $\mathbb{C}^m$. The Cartan generators therefore do not distinguish the copies. Splitting such copies requires an operator acting non-trivially on $\mathbb{C}^m$. For $p = 5$, Section 3 shows that only K_3 and K_4 can carry spectral information not fixed by $\hat{n}$ and $\hat{C}_2$. In the $(5, 4)$, $R = 4$ example, $\hat{n}$ and $\hat{C}_2$ have the same values on the two equivalent irreducible components, while the combination $-K_3 + K_4$ in $\tilde{H}$ produces the observed splitting. Whether the model admits a family of $\text{SU}(N)$ -invariant conserved operators, beyond functions of H , that is defined for all $N \geq 3$ and resolves these multiplicities remains open.

2.3 Cohomological structure

We first prove the trace identities needed for the integral normalization of Q_p and for the exact decoupling of the trace fermion in Section 2.4.

Proposition 1 (Traces of powers of an anticommuting matrix). *Let X be an $N \times N$ matrix whose entries are odd generators of an exterior algebra. Then*

$$\mathrm{Tr}(X^p) = \begin{cases} 0, & p \text{ even}, \\ p \tilde{T}_p(X), & p \text{ odd}, \end{cases} \quad (10)$$

where $\tilde{T}_p(X)$ is integral in the ordered monomial basis of the exterior algebra.

Proof. A nonzero term in $\mathrm{Tr}(X^p) = \sum_{i_0, \dots, i_{p-1}} X_{i_0 i_1} X_{i_1 i_2} \cdots X_{i_{p-1} i_0}$ corresponds to an ordered tuple of p distinct matrix entries $e_k = (i_k, i_{k+1})$, with the indices read modulo p . Any repeated entry gives a vanishing wedge product. The cyclic rotation $C_p : (e_0, e_1, \dots, e_{p-1}) \mapsto (e_1, \dots, e_{p-1}, e_0)$ acts on the set of such tuples. If a nontrivial rotation C_p^d ($0 < d < p$) fixed a tuple, then $e_0 = e_d$, contradicting distinctness. Thus every nonzero orbit has size exactly p . Within an orbit, a single rotation permutes p odd generators cyclically, contributing a sign $(-1)^{p-1}$. For even p , this gives alternating signs whose sum is zero. For odd p , all p terms in each orbit have the same sign, so the orbit sum is p times a single ordered monomial with integer coefficient. $\square$

Remark 1. The even case recovers the classical identity $\mathrm{Tr}(X^{2k}) = 0$ for matrices with anticommuting entries [35]. The same cyclic orbit argument proves both statements. No use of invariant theory is required.

The vanishing of the even traces uses only that the matrix entries are odd elements of an exterior algebra. It does not require independent generators and therefore applies to the traceless matrix A as well.

For odd p , Proposition 1 applied to Ψ gives the integral reduction

$$Q_p = \mathrm{Tr}(\Psi^p) = p \tilde{Q}_p. \quad (11)$$

Multiplication by $\tilde{Q}_p$ has sector matrices $\tilde{Q}_{p,R}$ with integer entries in the ordered bases of matrix unit monomials. Since $\tilde{Q}_p$ is nilpotent (see below), these maps define a cochain complex over $\mathbb{Z}$.

The next structural fact is that the BPS problem is cohomological for every odd p , not just for $p = 3$.

Proposition 2. *For every odd p , the supercharge $Q_p = \mathrm{Tr}(\Psi^p)$ satisfies $Q_p^2 = 0$.*

Proof. The operator Q_p is a sum of monomials, each of which is a product of exactly p of the mutually anticommuting creation operators Ψ_{ij} . The square Q_p^2 is therefore a sum over pairs of such monomials m_1, m_2 . Terms with $m_1 = m_2$ vanish because the square of an odd monomial is zero. For $m_1 \neq m_2$, both monomials have odd degree, so

$$m_1 m_2 = (-1)^{p^2} m_2 m_1 = -m_2 m_1.$$

The terms associated with (m_1, m_2) and (m_2, m_1) cancel. Thus $Q_p^2 = 0$ over every field of characteristic different from two. The argument uses only the odd degree of Q_p , not its trace form. The same cancellation is an integral identity. Since $Q_p^2 = p^2 \tilde{Q}_p^2 = 0$ and the integral exterior algebra has no torsion, $\tilde{Q}_p^2 = 0$ over $\mathbb{Z}$. $\square$

The integral normalization gives an overall factor p^2 in the Hamiltonian:

$$H = \{Q_p, Q_p^\dagger\} = p^2 \tilde{H}, \quad \tilde{H} := \{\tilde{Q}_p, \tilde{Q}_p^\dagger\}. \quad (12)$$

The numerical energies below are consistent with this normalization.

Since $Q_p^2 = 0$, the operator Q_p is a differential. Within each residue class $R \bmod p$, the sectors $V_R \xrightarrow{Q_R} V_{R+p} \xrightarrow{Q_{R+p}} \dots$ form a cochain complex. The BPS multiplicity in sector R is the dimension of the cohomology in degree R ,

$$h_R = \dim V_R - \text{rank } Q_R - \text{rank } Q_{R-p}. \quad (13)$$

We write

$$r_R := \text{rank } Q_R. \quad (14)$$

Thus the multiplicities at zero energy can be obtained from the ranks of the rectangular maps (4), without diagonalizing the full Hamiltonian.

Wedging elements of complementary degree gives a pairing in the highest exterior degree and the following symmetry of the sector ranks. Here $[c]_{\Lambda^{N^2}}$ denotes the coefficient of the ordered top form in c .

Theorem 1 (Symmetry of the sector ranks). *For every odd p with $p \leq 2N - 1$, the pairing in the highest exterior degree $\langle a, b \rangle = [a \wedge b]_{\Lambda^{N^2}}$ satisfies*

$$\langle Q_R a, b \rangle = (-1)^{pR} \langle a, Q_{N^2-p-R} b \rangle$$

for all $a \in V_R$, $b \in V_{N^2-p-R}$. Since the pairing is nondegenerate, Q_R and Q_{N^2-p-R} have equal rank:

$$r_R = r_{N^2-p-R}. \quad (15)$$

Consequently the polynomial $\mathcal{R}_{p,N}(x) = \sum_R r_R x^R$ is palindromic of degree $N^2 - p$.

Proof. Writing q for the element of degree p that acts by wedge multiplication, we have

$$\begin{aligned} \langle Q_R a, b \rangle &= [q \wedge a \wedge b]_{\Lambda^{N^2}} \\ &= (-1)^{pR} [a \wedge q \wedge b]_{\Lambda^{N^2}} \\ &= (-1)^{pR} \langle a, Q_{N^2-p-R} b \rangle. \end{aligned}$$

The second line moves q past the element a , which has degree R . The pairing $V_R \otimes V_{N^2-R} \rightarrow \mathbb{C}$ is nondegenerate, so the adjoint relation forces $\text{rank } Q_R = \text{rank } Q_{N^2-p-R}$. For $p \leq 2N - 1$, the description of the invariant generators in Appendix A.3 gives $Q_p \neq 0$. Since $\dim V_0 = 1$, the restriction Q_0 has rank one. Equation (15) then gives $r_{N^2-p} = 1$. For $R > N^2 - p$, the target of Q_R is zero. Hence $\mathcal{R}_{p,N}(x)$ has degree exactly $N^2 - p$. $\square$

Appendix A.3 compares this complex with the Chevalley–Eilenberg and Aomoto differentials [31, 36, 37]. It also explains why the present differential is not the usual Lie algebra differential.

2.4 $U(1)$ decoupling and the condition $p \leq 2N - 1$

Theorem 2 (Exact $U(1)$ decoupling). *With the orthogonal decomposition (2), the $U(1)$ trace fermion χ decouples exactly for every odd $p \geq 3$:*

$$Q_p = \text{Tr}(\Psi^p) = \text{Tr}(A^p). \quad (16)$$

Consequently the cochain complex splits as $\Lambda^\bullet(\mathbb{C}T_0) \otimes (\Lambda^\bullet(W_0), Q_p)$, and the BPS generating function acquires one exact factor of $(1+x)$ from the free χ mode. Any further factor of $1+x$ comes from the traceless complex.

Proof. Set $X := (\chi/\sqrt{N})\mathbf{1}$. Since χ is an odd Grassmann generator, $X^2 = 0$. Since χ and each A_{ij} are both odd, they anticommute: $XA = -AX$. Using $AX = -XA$ repeatedly gives $A^j X = (-1)^j X A^j$, so the terms containing one factor of X in the expansion of $(A + X)^p$ sum to

$$\sum_{j=0}^{p-1} A^j X A^{p-1-j} = \left(\sum_{j=0}^{p-1} (-1)^j \right) X A^{p-1} = X A^{p-1},$$

where the last equality uses $\sum_{j=0}^{p-1} (-1)^j = 1$ for odd p . Terms containing at least two factors of X vanish after the factors are anticommutated next to one another, since $X^2 = 0$. Hence $(A + X)^p = A^p + X A^{p-1}$. Taking the trace: $\text{Tr}(\Psi^p) = \text{Tr}(A^p) + \frac{\chi}{\sqrt{N}} \text{Tr}(A^{p-1})$. Since $p - 1$ is even, $\text{Tr}(A^{p-1}) = 0$ by Proposition 1, giving (16). The complex then splits as a tensor product with the free mode χ . By the Künneth formula [38], each traceless cohomology class occurs once with this mode empty and once with it occupied, giving the factor $(1 + x)$. $\square$

Corollary 3 (Factors implied by the complex). *For every odd $p \geq 3$ and every $N \geq 2$,*

$$(1 + x)^2 \mid Z_{\text{BPS}}^{(p,N)}(x). \quad (17)$$

If N is odd, then

$$(1 + x)^3 \mid Z_{\text{BPS}}^{(p,N)}(x). \quad (18)$$

The corresponding polynomial for the map ranks,

$$\mathcal{R}_{p,N}(x) = \sum_R \text{rank } Q_R x^R$$

satisfies

$$\mathcal{R}_{p,N}(x) = (1 + x) \mathcal{R}_{p,N}^0(x), \quad (19)$$

where $\mathcal{R}_{p,N}^0$ is the corresponding polynomial for the traceless complex. If N is odd and $p \leq 2N - 1$, then $(1 + x)^2 \mid \mathcal{R}_{p,N}(x)$.

Proof. Let

$$Z_0(x) = \sum_R \dim H^R(\Lambda^\bullet W_0, Q_p) x^R$$

be the generating polynomial for the traceless complex. Theorem 2 gives $Z_{\text{BPS}}(x) = (1 + x)Z_0$. Since Q_p has odd degree, the value $Z_0(-1)$ is the Euler characteristic of the traceless complex. Hence

$$Z_0(-1) = \sum_R (-1)^R \binom{N^2 - 1}{R} = (1 - 1)^{N^2 - 1} = 0.$$

This proves (17).

The pairing in the highest exterior degree, applied to $\Lambda^\bullet W_0$, gives $Z_0(x) = x^{N^2 - 1} Z_0(x^{-1})$. If N is odd, the exponent $N^2 - 1$ is even. For a polynomial with this symmetry and an even exponent, a zero at $x = -1$ has even multiplicity. Thus $(1 + x)^2 \mid Z_0$, which proves (18).

In degree R , the trace splitting gives

$$\Lambda^R(W_0 \oplus \mathbb{C}T_0) = \Lambda^R W_0 \oplus (\chi \wedge \Lambda^{R-1} W_0).$$

The differential has the two blocks Q_R^0 and $-Q_{R-1}^0$. Writing $s_R = \text{rank } Q_R^0$, we obtain $\text{rank } Q_R = s_R + s_{R-1}$, which proves (19). When N is odd, rank symmetry makes $\mathcal{R}_{p,N}$ reciprocal of even degree $N^2 - p$. Its zero at -1 therefore has even multiplicity, which supplies the second rank factor. $\square$

The supercharge is nonzero only when $p \leq 2N - 1$. For $p > 2N - 1$, the trace $\text{Tr}(A^p)$ vanishes identically [35]. Thus $p = 3, 5, 7$ first occur at $N = 2, 3, 4$, respectively. In particular, $Q_7 = 0$ at $N = 3$, so every state is BPS. Appendix A.3 explains the same condition in terms of primitive cocycles [31, 32].

When the cochain description is used, each residue class $a \in \mathbb{Z}/p\mathbb{Z}$ has Euler characteristic

$$I_a(N^2) = \sum_{R \equiv a \pmod{p}} (-1)^R \binom{N^2}{R} = \frac{1}{p} \sum_{\ell=0}^{p-1} \omega^{-a\ell} (1 - \omega^\ell)^{N^2}, \quad \omega = e^{2\pi i/p}. \quad (20)$$

These are the Witten indices of the p complexes obtained by fixing fermion number modulo p . They count BPS states with signs and can be computed from binomial coefficients without knowing the spectrum. Since p is odd, $(-1)^{R+p} = -(-1)^R$. On summing (13) over $R \equiv a \pmod{p}$ with weights $(-1)^R$, each rank Q_R occurs twice with opposite signs and cancels. Hence $I_a(N^2) = \sum_{R \equiv a \pmod{p}} (-1)^R h_R$. Their absolute values give an exact lower bound on the total BPS count,

$$\sum_R h_R \geq \sum_{a=0}^{p-1} |I_a|. \quad (21)$$

At fixed N , we record the difference between the computed count and this bound. This notion is separate from fortuity under changes of rank.

Equation (20) determines this bound whenever $p \leq 2N - 1$. For fixed p , at large N it grows exponentially in N^2 with rate $\log(2 \cos(\pi/(2p)))$. It therefore gives a rigorous lower bound on Z_{BPS} with the same rate. The derivation is given in Section 6.2.

3 Hamiltonian after normal ordering

The cohomological description determines the multiplicities at zero energy without displaying the interactions in the Hamiltonian. We now expand the reduced Hamiltonian after normal ordering. We first derive a formula valid for every odd p . We then obtain K_0, K_1, K_2 for $p = 5$ and every $N \geq 3$, and compare the commuting terms at $N = 3$ with the nonzero commutators found at $N = 4$.

3.1 General formula for odd p

The reduced supercharge creates from the vacuum an invariant state of p fermions under $\text{SU}(N)$,

$$|\Omega_p\rangle := \tilde{Q}_p |0\rangle.$$

For any state $|\Phi\rangle$ in the Fock space,

$$\langle \Phi | \tilde{H} | \Phi \rangle = \|\tilde{Q}_p |\Phi\rangle\|^2 + \|\tilde{Q}_p^\dagger |\Phi\rangle\|^2. \quad (22)$$

Thus the Hamiltonian measures whether this invariant state of p fermions can be added to or removed from $|\Phi\rangle$. A BPS state allows neither process.

We now separate these processes by the number of creation and annihilation operators. A term labelled by k contains k of each. As in supersymmetric SYK models whose supercharge is built from monomials of degree p in the fermions [39], the largest possible interaction contains $2p - 2$ fermion operators. Here the couplings are fixed by the trace and by $\text{SU}(N)$ symmetry rather than chosen randomly. For standard conventions in second quantization and normal ordering, see Ref. [40].

Let $W = \text{Mat}_N(\mathbb{C})$ with the Hilbert–Schmidt inner product. Use the orthonormal Hermitian basis $\{T_0, T_1, \dots, T_{N^2-1}\}$ fixed in Section 2.1, with $T_0 = \mathbf{1}/\sqrt{N}$ and $T_\alpha \in W_0$, and write

$$\Psi = \sum_{\mu=0}^{N^2-1} a_\mu^\dagger T_\mu, \quad a_0^\dagger = \chi, \quad a_\alpha^\dagger = b_\alpha^\dagger, \quad a_\mu = (a_\mu^\dagger)^\dagger, \quad \{a_\mu, a_\nu^\dagger\} = \delta_{\mu\nu}.$$

The invariant complex bilinear trace pairing $B(X, Y) = \text{Tr}(XY)$ also satisfies $B(T_\mu, T_\nu) = \delta_{\mu\nu}$. It therefore identifies W with W^* equivariantly under conjugation by $\text{SU}(N)$. We use this identification below. If $I = \{i_1 < \dots < i_k\}$, define

$$a_I^\dagger := a_{i_1}^\dagger \cdots a_{i_k}^\dagger, \quad a_I := (a_I^\dagger)^\dagger = a_{i_k} \cdots a_{i_1}.$$

By Theorem 2, the reduced supercharge of (11) is

$$\tilde{Q}_p = \frac{1}{p} \text{Tr}(A^p) = \sum_{|S|=p} (\Omega_p)_S a_S^\dagger. \quad (23)$$

In the chosen Hermitian basis, complex conjugation reverses the order inside each trace. Therefore

$$\overline{(\Omega_p)_S} = (-1)^{p(p-1)/2} (\Omega_p)_S.$$

Thus the components are real for $p \equiv 1 \pmod{4}$ and purely imaginary for $p \equiv 3 \pmod{4}$. Under the identification defined by the trace pairing, the coefficients $(\Omega_p)_S$ are the components of an $\text{SU}(N)$ -invariant form $\Omega_p \in \Lambda^p W^*$. Exact decoupling of the trace mode implies that this form is supported on W_0 .

Equip the exterior powers of W and W^* with their induced Hermitian inner products. The following map records the $(p-k)$ -particle wavefunction left after removing k particles from $|\Omega_p\rangle$:

$$\mathcal{A}_k : \Lambda^k W \longrightarrow \Lambda^{p-k} W^*, \quad \mathcal{A}_k(\alpha) = \iota_\alpha \Omega_p, \quad 0 \leq k \leq p. \quad (24)$$

Here $\iota_\alpha \Omega_p$ denotes contraction by the multivector α . Thus, if $\alpha = v_1 \wedge \dots \wedge v_k$, then $(\iota_\alpha \Omega_p)(w_1, \dots, w_{p-k}) = \Omega_p(v_1, \dots, v_k, w_1, \dots, w_{p-k})$, extended linearly. For disjoint index subsets I, T , each written in increasing order, the matrix elements of $\mathcal{A}_k$ are

$$(\mathcal{A}_k)_{T,I} = \epsilon(T, I) (\Omega_p)_{T \cup I}, \quad |I| = k, \quad |T| = p - k,$$

where $\epsilon(T, I)$ is the sign that orders the concatenated wedge product. The form Ω_p vanishes whenever one of its arguments lies in the trace direction. The common phase of its components implies that $\mathcal{A}_k^\dagger \mathcal{A}_k$ is real symmetric.

The contraction matrices have a direct interpretation in terms of correlations. With the ordering of subsets fixed above, define

$$\Gamma_{I,J}^{(k)} := \langle \Omega_p | a_I^\dagger a_J | \Omega_p \rangle = (\mathcal{A}_k^\dagger \mathcal{A}_k)_{I,J}. \quad (25)$$

Thus $\Gamma^{(k)}$ is the unnormalized reduced density matrix for k particles in $|\Omega_p\rangle$, in the index convention of (25) [41], and

$$\text{Tr} \Gamma^{(k)} = \binom{p}{k} \|\Omega_p\|^2. \quad (26)$$

The operator K_k applies this correlation matrix to the full Fock space:

$$K_k := \sum_{\substack{|I|=k \\ |J|=k}} (\mathcal{A}_k^\dagger \mathcal{A}_k)_{I,J} a_I^\dagger a_J. \quad (27)$$

The strings $a_I^\dagger a_J$, with $|I| = |J|$, form a basis of operators after normal ordering that preserve fermion number. The expansion is therefore complete. In particular, K_k is the contribution with k creation and k annihilation operators.

For a single monomial the alternating structure has a simple interpretation in terms of Pauli blocking. If $q = a_1^\dagger \cdots a_p^\dagger$ and $n_i = a_i^\dagger a_i$, then

$$\{q, q^\dagger\} = \prod_{i=1}^p n_i + \prod_{i=1}^p (1 - n_i). \quad (28)$$

The two terms project onto the configurations in which all p modes are occupied or all are empty. Expanding the second product gives an alternating sum in the occupied modes. For odd p , its term of degree p cancels the first product. The following proposition is the corresponding statement for the supercharge Q_p defined by one trace.

Proposition 4 (Hamiltonian after normal ordering). *For every odd p ,*

$$\tilde{H} = \{\tilde{Q}_p, \tilde{Q}_p^\dagger\} = \sum_{k=0}^{p-1} (-1)^k K_k. \quad (29)$$

Each K_k is positive semidefinite, preserves fermion number, and is separately $\text{SU}(N)$ -invariant. Equivalently, $H = p^2 \sum_{k=0}^{p-1} (-1)^k K_k$.

Proof. For $0 \leq k \leq p-1$, only the normal ordering of $\tilde{Q}_p^\dagger \tilde{Q}_p$ contributes a term with k creation and k annihilation operators. Choosing the $p-k$ contracted modes and moving the k remaining creation operators past the k remaining annihilation operators gives $(-1)^{k^2} = (-1)^k$. Hence, for $|I| = |J| = k$,

$$\begin{aligned} \text{coeff}_{a_I^\dagger a_J} \tilde{H} &= (-1)^{k^2} \sum_{\substack{|T|=p-k \\ T \cap (I \cup J) = \emptyset}} \overline{\epsilon(T, J)(\Omega_p)_{T \cup J}} \epsilon(T, I)(\Omega_p)_{T \cup I} \\ &= (-1)^k (\mathcal{A}_k^\dagger \mathcal{A}_k)_{J, I} = (-1)^k (\mathcal{A}_k^\dagger \mathcal{A}_k)_{I, J}, \quad 0 \leq k \leq p-1. \end{aligned}$$

The last equality uses the real symmetry proved above. For $k = p$, the term from $\tilde{Q}_p^\dagger \tilde{Q}_p$ is $(-1)^{p^2}$ times the term from $\tilde{Q}_p \tilde{Q}_p^\dagger$, so the two cancel because p is odd. This proves (29). The Gram form makes each K_k positive semidefinite, and invariance follows from that of Ω_p . $\square$

3.2 The quintic Hamiltonian

In the quintic model, K_0 is the constant term. The operators K_1 and K_2 contain respectively one and two creation operators, together with the same number of annihilation operators, in the expansion obtained by normal ordering $\tilde{H}$. All three are fixed analytically by $\text{SU}(N)$ representation theory, normalization, and the contraction identities for the primitive form. Let $\hat{C}_2$ be the quadratic Casimir for the conjugation action of $\text{SU}(N)$ on the full Fock space. The trace mode is an $\text{SU}(N)$ singlet, and $\hat{C}_2$ is normalized to have eigenvalue N on the adjoint representation. For $p = 5$ and every $N \geq 3$, the following identities hold as operators on the full Fock space:

$$K_0 = \frac{N(N^2 - 1)(N^2 - 4)}{5}, \quad (30)$$

$$K_1 = N(N^2 - 4) \hat{n}, \quad (31)$$

$$K_2 = N\hat{n}(\hat{n} - 1) + (N^2 - 6)(N\hat{n} - \hat{C}_2). \quad (32)$$

All adjoints and norms below use the Fock inner product induced by the Hilbert–Schmidt inner product on W . The scale of the invariant form is fixed by

$$\tilde{Q}_5 = \frac{1}{5} \text{Tr}(A^5) = \frac{1}{5} \text{Tr}(\Psi^5).$$

Its coefficients in the original wedge basis built from matrix units are integers by (11). For comparison with Ref. [42], set $\lambda_a = \sqrt{2} T_a$, so that $\text{Tr}(\lambda_a \lambda_b) = 2\delta_{ab}$, as in that reference. Let f_{abc} be the real structure constants defined by $[\lambda_a, \lambda_b] = 2if_{abc}\lambda_c$, equivalently $[T_a, T_b] = i\sqrt{2}f_{abc}T_c$. We use normalized antisymmetrization,

$$T_{[a_1 \dots a_5]} := \frac{1}{5!} \sum_{\sigma \in S_5} \text{sgn}(\sigma) T_{a_{\sigma(1)}} \cdots T_{a_{\sigma(5)}}.$$

Let Ω^{lit} denote the invariant cocycle of degree five in the normalization used in Ref. [42]. With this convention, equation (117) of that reference gives $\text{Tr} \lambda_{[a_1 \dots a_5]} = -2\Omega_{a_1 \dots a_5}^{\text{lit}}$. Collecting terms in $\text{Tr}(A^5)/5$ into increasing index sets therefore gives

$$(\Omega_5)_{a_1 \dots a_5} = \frac{5!}{5} \text{Tr} T_{[a_1 \dots a_5]} = -6\sqrt{2} \Omega_{a_1 \dots a_5}^{\text{lit}}, \quad a_1 < \dots < a_5. \quad (33)$$

After this normalization, equations (73), (76), (82), and (84) of that reference give

$$\mathcal{A}_1^\dagger \mathcal{A}_1 = N(N^2 - 4) \mathbf{1}_{\text{adj}}, \quad (34)$$

$$\mathcal{A}_2^\dagger \mathcal{A}_2 = N(N^2 - 4) P_{\text{adj}} + 2N P_\perp. \quad (35)$$

The first identity holds on W_0 , and the second on $\Lambda^2 W_0$. Here P_{adj} projects onto the adjoint summand and $P_\perp$ onto its orthogonal complement [33, 42]. In translating equation (84), we used $(P_{\text{adj}})_{ab,cd} = 2N^{-1} f_{abe} f_{cde}$ in the wedge basis indexed by increasing pairs $a < b$ and $c < d$. The first identity gives (31). Since $K_0 = \|\Omega_5\|^2$ and $\text{Tr}(\mathcal{A}_1^\dagger \mathcal{A}_1) = 5\|\Omega_5\|^2$, taking the trace gives (30). For $N \geq 3$, the orthogonal complement of the adjoint representation in $\Lambda^2 W_0$ is a conjugate pair of irreducible representations. Indeed, writing $W = V \otimes V^*$ with $V = \mathbb{C}^N$, one has

$$\Lambda^2(V \otimes V^*) \simeq (S^2 V \otimes \Lambda^2 V^*) \oplus (\Lambda^2 V \otimes S^2 V^*) \simeq 2 \text{adj} \oplus V_\lambda \oplus \overline{V}_\lambda,$$

where V_λ is the irreducible representation with highest weight $\lambda = (2, 0, \dots, 0, -1, -1)$, written in the standard N -tuple notation, and $\overline{V}_\lambda$ is its conjugate. Since $V \otimes V^* \simeq \mathbb{C} \oplus W_0$, it follows that

$$\Lambda^2 W_0 \simeq \text{adj} \oplus V_\lambda \oplus \overline{V}_\lambda.$$

For a highest weight $\ell = (\ell_1, \dots, \ell_N)$, with $\sum_i \ell_i = 0$, the quadratic Casimir in the present normalization is

$$C_2(\ell) = \frac{1}{2} \sum_{i=1}^N \ell_i (\ell_i + N + 1 - 2i).$$

This gives N for the adjoint weight $(1, 0, \dots, 0, -1)$ and $2N$ for λ and its conjugate. Since $W = W_0 \oplus \mathbb{C} T_0$,

$$\Lambda^2 W = \Lambda^2 W_0 \oplus (T_0 \wedge W_0).$$

On $\Lambda^2 W_0$, $\hat{n} = 2$, and the right side of (32) has eigenvalues $N(N^2 - 4)$ and $2N$ on the adjoint and its complement. On $T_0 \wedge W_0$, $\mathcal{A}_2^\dagger \mathcal{A}_2 = 0$. There $\hat{n} = 1$ and $\hat{C}_2 = N$, so the right side also vanishes. Each generator of the conjugation action contains one creation and one annihilation operator. Since $\hat{C}_2$ is quadratic in these generators, its expansion after normal ordering contains no term with more than two creation and two annihilation operators. After normal ordering, $\hat{n}(\hat{n} - 1)$ contains only terms with two creation and two annihilation operators. On the traceless sector with one fermion, $\hat{C}_2 = N\mathbf{1}$, while it vanishes on the trace mode. Its part with one creation and one annihilation operator is therefore $N\hat{n}$, so $N\hat{n} - \hat{C}_2$ also contains only terms with two creation and two annihilation operators. For any such operator, its matrix elements in the ordered basis $a_I^\dagger |0\rangle$ of $\Lambda^2 W$ are exactly the coefficients multiplying $a_I^\dagger a_J$. Its action on $\Lambda^2 W$ therefore determines the operator on the full Fock space. Since the two sides of (32) agree on both summands of $\Lambda^2 W$, the identity follows.

Thus the contribution for one particle treats every traceless mode equally. For a pair, the strength is $N(N^2 - 4)$ in the adjoint channel and $2N$ in its complement. These are exact operator identities for every $N \geq 3$.

It follows that

$$\tilde{H}_{\leq 2} := K_0 - K_1 + K_2 = \frac{N(N^2 - 1)(N^2 - 4)}{5} + N\hat{n}(\hat{n} - 3) - (N^2 - 6)\hat{C}_2. \quad (36)$$

Because Ω_p vanishes on the trace direction, $\mathcal{A}_k^\dagger \mathcal{A}_k$ is supported on $\Lambda^k W_0$. Hence K_k commutes with n_χ . Together with conservation of the total fermion number, this gives $[K_k, \hat{n}] = 0$. Its $SU(N)$ -invariance also gives $[K_k, \hat{C}_2] = 0$. For $p = 5$ and every $N \geq 3$, the identities (30)–(32) and the commutation relations above show that each of K_0 , K_1 and K_2 commutes with every K_j . Their combination $\tilde{H}_{\leq 2}$ therefore also commutes with every K_j . Consequently, the only commutator among the five terms that can be nonzero is $[K_3, K_4]$. For $p = 5$,

$$\tilde{H} = \tilde{H}_{\leq 2} - K_3 + K_4.$$

Thus only K_3 and K_4 can contain spectral information not fixed by the traceless fermion number and the quadratic Casimir. At fixed $\hat{n}$, $\tilde{H}_{\leq 2}$ is scalar on each $SU(N)$ irreducible representation. An invariant term with more creation and annihilation operators can nevertheless act on the multiplicity space when the same representation occurs more than once. This is how K_3 and K_4 can split states with identical $(\hat{n}, \hat{C}_2)$.

For comparison, consider the cubic model. Write $K_k^{(3)}$ for the corresponding cubic operators. For $a < b < c$, the cubic form has components

$$(\Omega_3)_{abc} = \text{Tr}(T_a[T_b, T_c]) = i\sqrt{2} f_{abc}.$$

The identity $f_{acd}f_{bcd} = N\delta_{ab}$ therefore gives

$$(\mathcal{A}_1^{(3)})^\dagger \mathcal{A}_1^{(3)} = N\mathbf{1}_{\text{adj}}.$$

This gives $K_1^{(3)} = N\hat{n}$. Taking the trace and using (26) gives $K_0^{(3)} = \|\Omega_3\|^2 = N(N^2 - 1)/3$. On $\Lambda^2 W_0$, the same identity for the structure constants gives

$$(\mathcal{A}_2^{(3)})^\dagger \mathcal{A}_2^{(3)} = NP_{\text{adj}} = 2N\mathbf{1} - \hat{C}_2.$$

On $T_0 \wedge W_0$, both $(\mathcal{A}_2^{(3)})^\dagger \mathcal{A}_2^{(3)}$ and $N\hat{n} - \hat{C}_2$ vanish. Both $K_2^{(3)}$ and $N\hat{n} - \hat{C}_2$ contain only terms with two creation and two annihilation operators, so their agreement on the two summands of $\Lambda^2 W$ gives $K_2^{(3)} = N\hat{n} - \hat{C}_2$ on the full Fock space. Thus

$$K_0^{(3)} = \frac{N(N^2 - 1)}{3}, \quad K_1^{(3)} = N\hat{n}, \quad K_2^{(3)} = N\hat{n} - \hat{C}_2. \quad (37)$$

Normal ordering stops after the terms with two creation and two annihilation operators, and the contributions from the fermion number operator cancel:

$$H_3 = 9(K_0^{(3)} - K_1^{(3)} + K_2^{(3)}) = 3N(N^2 - 1) - 9\hat{C}_2. \quad (38)$$

This recovers (9). The cubic Hamiltonian is determined by the quadratic Casimir because the $\hat{n}$ terms in $K_1^{(3)}$ and $K_2^{(3)}$ cancel, and no $K_k^{(3)}$ with $k > 2$ appears in H_3 .

Since K_k annihilates states with fewer than k traceless fermions, the reduced quintic Hamiltonian satisfies $\tilde{H} = \tilde{H}_{\leq 2}$ whenever $\hat{n} \leq 2$. As a check, at $(5, 3)$ there are no zero modes in V_0 or V_1 , while

$$V_2 = (\mathbf{10} \oplus \overline{\mathbf{10}}) \oplus 2 \cdot \mathbf{8}, \quad H|_{V_2} = 25(18 - 3\hat{C}_2).$$

Here one $\mathbf{8}$ lies in $\Lambda^2 W_0$, where $\hat{n} = 2$, and the other lies in $T_0 \wedge W_0$, where $\hat{n} = 1$. The displayed formula applies to both because $\hat{n}(\hat{n} - 3) = -2$ for either value. The relevant Casimir eigenvalues are $\hat{C}_2(\mathbf{8}) = 3$ and $\hat{C}_2(\mathbf{10}) = \hat{C}_2(\overline{\mathbf{10}}) = 6$. This gives 20 BPS states and 16 states at $E = 225$, explaining the onset $h_2 = 20$ in (41).

3.3 Commutators at $N = 3$ and $N = 4$

Proposition 5 (Commutativity at $(5, 3)$). *For the quintic model at $N = 3$, the five operators $K_0, \dots, K_4$ commute pairwise on the full Fock space. This follows analytically from Hodge duality between the invariant cubic and quintic forms on $\mathfrak{su}(3)$.*

A detailed proof is given in Appendix A.2. Indeed, at $N = 3$ the traceless space has dimension eight. Its Hodge star maps forms of degree three to forms of degree five. The invariant forms obey $\Omega_5 = c * \Omega_3$ for some nonzero scalar c . Their squared norms in the present convention obey $\|\Omega_5\|^2 = 3\|\Omega_3\|^2$ [34, 42]. Equation (76) in Appendix A.2 shows that each quintic contraction matrix is $3\tilde{Q}_3^\dagger \tilde{Q}_3$ on $\Lambda^k W_0$. Normal ordering the cubic anticommutator and using (38) gives $\tilde{Q}_3^\dagger \tilde{Q}_3 = 8 - \hat{C}_2 - \tilde{Q}_3 \tilde{Q}_3^\dagger$. This converts $\tilde{Q}_3^\dagger \tilde{Q}_3$ into the $\tilde{Q}_3 \tilde{Q}_3^\dagger$ form used in the explicit formula there. That formula expresses every K_k through the commuting operators $\hat{n}$, $\hat{C}_2$, and $\tilde{Q}_3 \tilde{Q}_3^\dagger$. Thus the $(5, 3)$ commutativity follows from this Hodge duality at $N = 3$.

The same pairwise commutativity does not hold in the computed $N = 4$ examples. In the numerical $(5, 4)$, $R = 4$ example described in Section 4.2, the two 20-dimensional blocks have the same $\hat{n}$ and the same central Casimirs. In $\tilde{H}$, their splitting is therefore produced entirely by $-K_3 + K_4$, equivalently by $25(-K_3 + K_4)$ in H . The two operators commute in that sector, but a direct calculation with integer matrices gives $[K_3, K_4] \neq 0$ at $R = 5$. At $(7, 4)$, the same calculation gives $[K_3, K_4] \neq 0$, $[K_3, K_5] \neq 0$, and $[K_4, K_5] \neq 0$ in sector $R = 5$. Here $R = 5$ denotes the full sector V_5 , including both possible occupations of the trace mode.

The ancillary program `verify_normal_ordering.py` constructs the K_k , checks the operator identities used in Proposition 5, and reproduces these commutators using integer arithmetic. The nonzero commutators show that the displayed K_k cannot be simultaneously diagonalized in the $(5, 4)$ and $(7, 4)$ examples. They do not establish chaos or a failure of integrability, and they do not exclude another commuting family.

4 Numerical method and results

This section explains how the sector maps and Hamiltonian matrices are constructed. It also states which conclusions are analytic and which are inferred from numerical ranks.

4.1 Computational method and status of the data

We work in the ordered monomial basis of each V_R , so the matrices Q_R are sparse. Define

$$A_R = \begin{bmatrix} Q_R \\ Q_{R-p}^\dagger \end{bmatrix} : V_R \rightarrow V_{R+p} \oplus V_{R-p}. \quad (39)$$

Then

$$H_R = A_R^\dagger A_R, \quad (40)$$

so H_R and $A_R A_R^\dagger$ have the same nonzero eigenvalues. It is therefore more efficient to diagonalize $A_R A_R^\dagger$ whenever $\dim V_{R+p} + \dim V_{R-p} < \dim V_R$. For $p = 5$, $N = 4$, this is what makes the middle sectors feasible.

We imposed three checks on the data:

1. the zero multiplicities in each sector from (13) agree with those from H_R ,
2. the data obey the particle-hole symmetry (6),
3. for the full spectra with $N \leq 4$, the energy multiplicities in each sector sum to $\dim V_R$.

Scope of the numerical results. The results below concern only the stated pairs (p, N) and fall into three categories.

- (i) At the four complete cases $(5, 3)$, $(5, 4)$, $(5, 5)$, $(7, 4)$ we report a numerical BPS multiplicity h_R for every R .
- (ii) The ranks in each sector for all four cases are supplied for every R in the ancillary file `rank_data.json`. For $(5, 4)$, the spectrum of H is tabulated through half filling in Appendix B.1. The remaining sectors follow by particle-hole conjugation. For $(5, 3)$ and $(7, 4)$, we record the distinct energies together with the BPS multiplicities, with the complete spectra used only for the consistency checks listed above. At $(5, 5)$ only the multiplicities at zero energy were computed, since the nonzero levels would require diagonalizing the middle sectors.
- (iii) At $(7, 5)$ the data are partial. Only the first sectors $R \leq 6$ were computed, giving $q_{\min}$ but not the full spectrum.

The multiplicities are inferred from numerical ranks of explicitly constructed integer sector maps. For $N \leq 4$, the zero and nonzero clusters of singular values are separated by more than four orders of magnitude. In the $N = 5$ block computations, the corresponding eigenvalue clusters of the Gram matrices are separated by more than two orders of magnitude. The $N = 5$ results were also reproduced by the second implementation described in Appendix B.

As an independent check on the calculation with floating-point numbers, the ranks at $(5, 3)$, $(5, 4)$, and $(7, 4)$ were recomputed from the integer matrices of the reduced differential $\tilde{Q}_p$. Exact linear algebra was performed over the fields with characteristics $2^{31} - 1$, 2147483629, and p . All three modular ranks agree with one another and with the numerical rank in every sector. This is an exact check over those fields, but agreement over finitely many fields does not prove the rank over $\mathbf{Q}$. Accordingly, the positive BPS multiplicities below remain numerical unless stated otherwise. For an integer matrix, reduction modulo a prime cannot increase its rank. Thus a calculation with full column rank over one finite field proves injectivity over $\mathbf{Q}$. Here $\tilde{Q}_p$ over a finite field means the reduction of its integral sector matrices, not division by p in that field. If the cohomology in sector R vanishes over one finite field, the incoming and outgoing modular ranks add to $\dim V_R$. The corresponding ranks over $\mathbf{Q}$ are at least as large, while $\tilde{Q}_p^2 = 0$ prevents their sum from exceeding $\dim V_R$. Equality follows, so the cohomology over $\mathbf{Q}$ also vanishes. No such modular certification is presently claimed for the $N = 5$ sectors. The ancillary file `rank_data.json` lists the reported ranks, sector dimensions, nullities, and cohomology dimensions. The bundle also contains the program for the modular rank checks, the program that checks the normal ordering, and the scripts that generate the figures.

We now present the numerical BPS generating functions for the quintic model at $N = 3, 4, 5$ and the $p = 7$ model at $N = 4$. We also give the case $(7, 3)$, where $Q_7 = 0$, and partial data at $(7, 5)$. We then use projections to compare the cohomology at different values of N .

4.2 The quintic model: $p = 5$

The first nontrivial quintic case is $N = 3$, where the Fock space is small enough for complete diagonalization. At $N = 4$, Gram matrices are used in the middle sectors.

$N = 3$

The Hilbert space has dimension $2^9 = 512$. The numerical spectrum contains three distinct energies,

$$E \in \{0, 225, 600\},$$

and the BPS sector has dimension 440. The multiplicities at zero energy, listed by fermion number, are

$$(h_0, h_1, \dots, h_9) = (0, 0, 20, 75, 125, 125, 75, 20, 0, 0). \quad (41)$$

In particular, the first nonzero BPS sector is $R = 2$: $h_0 = h_1 = 0$.

$N = 4$

The Hilbert space has dimension $2^{16} = 65,536$. The numerical spectrum contains nine distinct energies,

$$E \in \{0, 225, 600, 900, 1100, 1400, 2100, 2400, 3600\},$$

and the total BPS multiplicity is

$$\sum_R h_R = 44,000. \quad (42)$$

The multiplicities at zero energy are

$$(h_0, h_1, \dots, h_{16}) = (0, 0, 0, 0, 125, 1375, 5000, 9625, 11750, 9625, 5000, 1375, 125, 0, 0, 0, 0). \quad (43)$$

Equivalently, the BPS spectrum begins at fermion number $R = 4$. The charge profiles for the four complete nontrivial cases are compared in Figure 3.

Equivalent representations at different energies. In sector $R = 4$, consider the subspace with an unoccupied trace mode. Writing $\mathcal{E}_E$ for the eigenspace with energy E in this subspace, the numerical $SU(4)$ decompositions are

$$\mathcal{E}_0 = [0, 4, 0] \oplus [0, 2, 0], \quad \mathcal{E}_{2100} = [1, 0, 1] \oplus [0, 2, 0].$$

The two eigenspaces have dimensions 125 and 35, respectively. Each contains one copy of the same 20-dimensional irreducible representation $[0, 2, 0]$. Since these two summands are equivalent, they have identical eigenvalues for every central Casimir [43], but they occur at different energies. No function of the central Casimirs can distinguish the two summands. Section 3 constructs a commuting family at $(5, 3)$. Whether the model more generally admits a family of pairwise commuting $SU(N)$ -invariant conserved operators, beyond functions of H , that is defined for all $N \geq 3$ and resolves such multiplicities remains open.

Numerical result 1 (Quintic BPS generating polynomials). The numerical BPS generating polynomials are

$$Z_{\text{BPS}}^{(5,3)}(x) = 5x^2(1+x)^3(4+3x+4x^2), \quad (44)$$

$$Z_{\text{BPS}}^{(5,4)}(x) = 125x^4(1+x)^4(1+7x+6x^2+7x^3+x^4). \quad (45)$$

Evaluating at $x = 1$ gives 440 and 44,000, respectively. After the displayed factors are removed, particle-hole symmetry gives the reciprocal polynomials

$$T_3^{(5)}(x) = 4 + 3x + 4x^2, \quad T_4^{(5)}(x) = 1 + 7x + 6x^2 + 7x^3 + x^4.$$

4.3 The $p = 7$ model

At $N = 3$, one has $Q_7 = 0$ because $7 > 2N - 1$. At $N = 4$, it provides a nontrivial calculation at a different value of p .

$N = 3$

Since $p = 7 > 2N - 1 = 5$, the trace in the supercharge vanishes. Accordingly $Q_7 \equiv 0$, the Hamiltonian vanishes identically, and the full Fock space is BPS:

$$Z_{\text{BPS}}^{(7,3)}(x) = (1+x)^9. \quad (46)$$

This agrees with the condition (80).

$N = 4$

At $N = 4$, the Hilbert space again has dimension 65,536. The numerical spectrum contains eight distinct energies,

$$E \in \{0, 1764, 4410, 7350, 7840, 14700, 18816, 35280\},$$

and the total BPS multiplicity is

$$\sum_R h_R = 59,136. \quad (47)$$

The multiplicities at zero energy, listed by fermion number, are

$$(h_0, h_1, \dots, h_{16}) = (0, 0, 0, 70, 847, 3395, 7518, 11319, 12838, 11319, 7518, 3395, 847, 70, 0, 0, 0). \quad (48)$$

Here the first nonzero BPS sector is $R = 3$.

Numerical result 2 (BPS generating polynomial for $p = 7$). For $p = 7$, $N = 4$, the numerical BPS generating polynomial is

$$Z_{\text{BPS}}^{(7,4)}(x) = 7x^3(1+x)^4(10 + 81x + 101x^2 + 144x^3 + 101x^4 + 81x^5 + 10x^6). \quad (49)$$

The polynomial

$$T_4^{(7)}(x) = 10 + 81x + 101x^2 + 144x^3 + 101x^4 + 81x^5 + 10x^6$$

is reciprocal, as follows from particle-hole symmetry. The BPS fraction is $59,136/65,536 \approx 90.2\%$, which is higher than in the quintic model at the same matrix size.

Figure 2 compares the distinct energy levels in the two computed $N = 4$ models. The representation comparison in the quintic case shows that the quadratic Casimir does not determine all these levels.

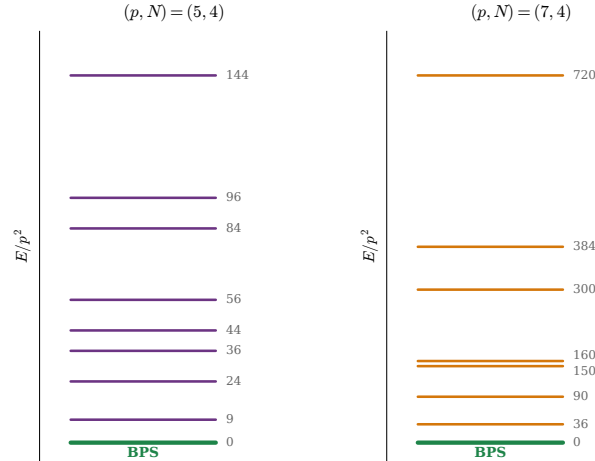


Figure 2: Distinct numerical energy levels in the calculated models. Each horizontal line marks a value of E/p^2 , and the green line is the BPS value $E = 0$. Equivalent irreducible representations can occur at different energies, so the quadratic Casimir does not determine these levels.

4.4 Rank sequences and the $N = 5$ data

The cohomology formula (13) determines h_R from the ranks of the rectangular maps Q_R . For $p = 5$, $N = 4$, the reported rank sequence is

$$(r_0, \dots, r_{11}) = (1, 16, 120, 560, 1695, 2992, 2992, 1695, 560, 120, 16, 1), \quad (50)$$

which is palindromic (as guaranteed by Theorem 1) and unimodal. The computation over finite fields certifies injectivity for $R \leq 3$, so there are no BPS states at those charges. Within the reported numerical rank assignment, the first rank deficiency occurs at $R = 4$, where $\binom{16}{4} - r_4 = 125$. This reported kernel dimension equals the power 5^3 that will appear in the factorization of the BPS polynomial in Section 5. The equality is not general: at $(5, 3)$, the first BPS multiplicity is $h_2 = 20$, whereas the common power dividing the reduced coefficients is 5.

At $N = 5$, the sector maps were computed using sparse matrix assembly and block decomposition of the Gram matrices $Q_R^\top Q_R$. Computational details are given in Appendix B. The results are summarized in Table 1.

R	$\dim V_R$	$\dim \ker Q_R$	$\text{rank } Q_{R-5}$	h_R
0-4	$\leq 12,650$	0	—	0
5	53,130	1	1	0
6	177,100	25	25	0
7	480,700	300	300	0
8	1,081,575	36,050	2,300	33,750
9	2,042,975	319,525	12,650	306,875
10	3,268,760	1,221,254	53,129	1,168,125

Table 1: Numerical BPS data at low fermion number for $(p, N) = (5, 5)$. For $R \leq 4$, the reported rank is full. The sectors $R = 11, 12$ follow from the symmetry $r_R = r_{N^2-p-R}$, and the remaining sectors follow from particle-hole symmetry.

Thus $q_{\min} = 8$ at $(p, N) = (5, 5)$. This is one above the empirical $N \leq 4$ extrapolation stated in Section 5. Figure 4 shows how the cohomology first becomes nonzero. The first two nonzero BPS multiplicities are

$$h_8^{(5,5)} = 33,750 = 2 \cdot 3^3 \cdot 5^4, \quad h_9^{(5,5)} = 306,875 = 5^4 \cdot 491. \quad (51)$$

Both reported values are divisible by 5^4 but not 5^5 , and their greatest common divisor is exactly $5^4 = 625$. By particle-hole symmetry, $h_{17}^{(5,5)} = 33,750$ and $h_{16}^{(5,5)} = 306,875$.

For the first two surviving sectors ($R = 8, 9$), the individual nullities and incoming ranks are each divisible by 5^2 but not 5^4 . For instance,

$$\dim \ker Q_8 = 36,050 = 5^2 \cdot 1,442, \quad \text{rank } Q_3 = 2,300 = 5^2 \cdot 92,$$

but $h_8 = 5^4 \cdot 54$ because $1,442 - 92 = 1,350 = 5^2 \cdot 54$. At $R = 10$, neither individual term is divisible by 5^2 , whereas their difference is divisible by 5^4 : $h_{10} = 1,168,125 = 5^4 \cdot 1,869$. Thus the extra 5^4 divisibility appears only after subtracting the incoming rank from the kernel dimension: $\dim \ker Q_R \equiv \text{rank } Q_{R-5} \pmod{5^4}$. It is not divisibility of either term separately by a higher power of p .

The numerical calculation at $R = 10$ gives $r_{10} = 2,047,506$ and $h_{10} = 1,168,125$. Together with the symmetry of the ranks and particle-hole symmetry, these rank assignments give the following numerical generating polynomial:

$$\begin{aligned} Z_{\text{BPS}}^{(5,5)}(x) = & 33,750x^8 + 306,875x^9 + 1,168,125x^{10} + 2,556,875x^{11} \\ & + 3,674,375x^{12} + 3,674,375x^{13} + 2,556,875x^{14} \\ & + 1,168,125x^{15} + 306,875x^{16} + 33,750x^{17}. \end{aligned} \quad (52)$$

The corresponding numerical total is 15,480,000. Section 5 shows that the same polynomial contains $(1+x)^5$ and that r_{10} can instead be determined from the ranks at lower fermion number and the exact relation $(1+x) \mid \mathcal{R}_{5,5}(x)$.

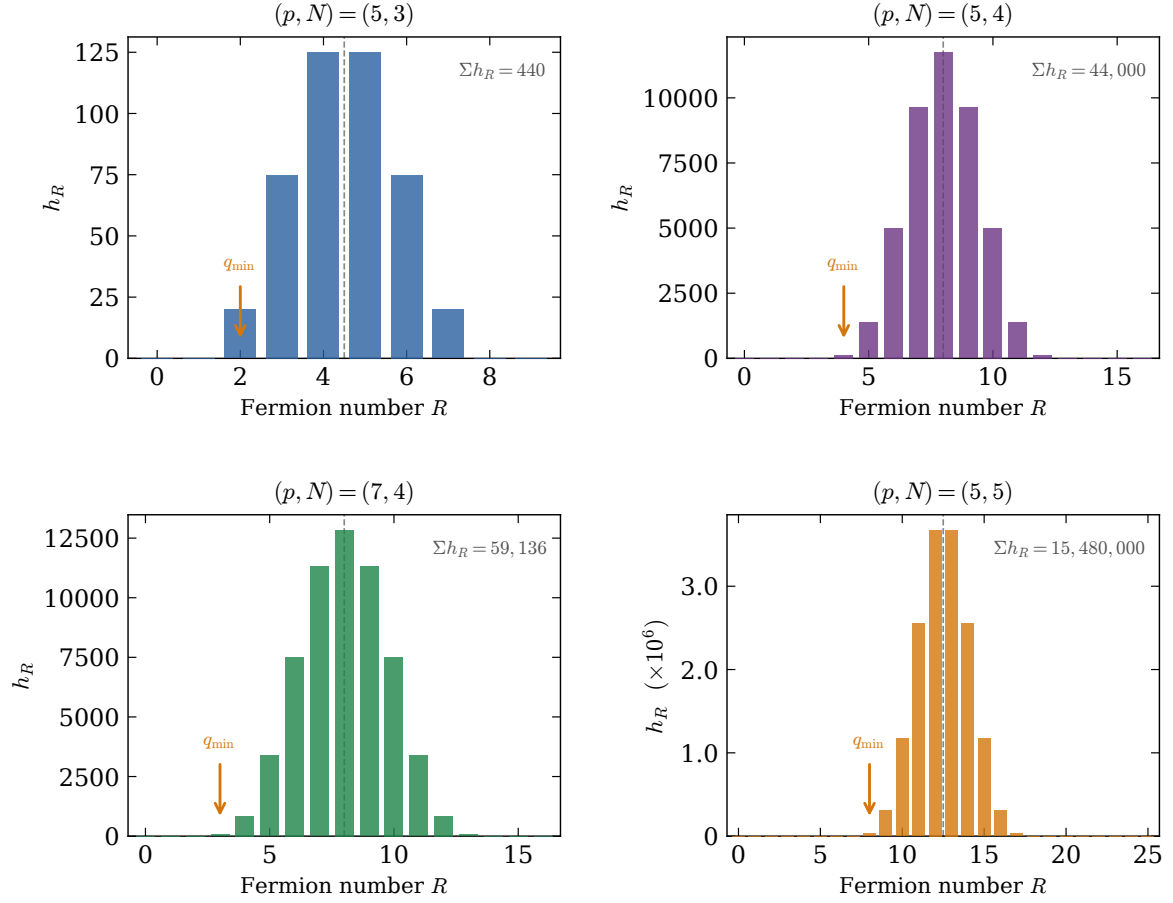


Figure 3: Computed BPS charge profiles. Numerical BPS multiplicities h_R , resolved by fermion number, for the four nontrivial cases with data in every sector. The dashed line marks the particle-hole symmetry axis at half filling ($R = N^2/2$). Arrows indicate the minimum BPS charge $q_{\min}$. At $(5, 5)$ it is one above the empirical $N \leq 4$ extrapolation in (61).

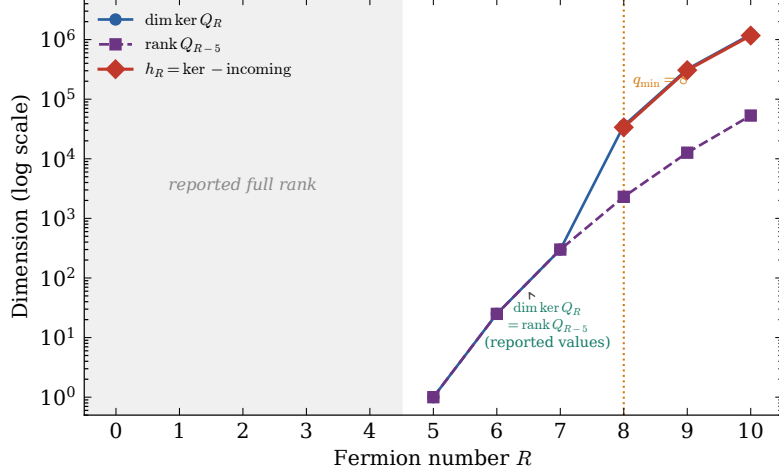


Figure 4: For the reported numerical ranks, the figure shows the onset of BPS cohomology at $(p, N) = (5, 5)$. For $R \leq 4$, Q_R is injective and no kernel exists. At $R = 5, 6, 7$, a kernel appears in blue, but its dimension equals the incoming rank in purple, so $h_R = 0$. Cohomology first survives at $R = 8 = q_{\min}$, where the kernel dimension exceeds the incoming rank. The red diamonds plot $h_R = \dim \ker Q_R - \text{rank } Q_{R-5}$.

For $p = 7$, $N = 5$, the same numerical method gives full column rank for Q_R at $R = 0, \dots, 5$ and

$$h_6^{(7,5)} = 2,352 = 2^4 \cdot 3 \cdot 7^2 \quad (53)$$

(with no incoming map since $R - p < 0$), so $q_{\min} = 6$. This is again one above the empirical extrapolation from $N \leq 4$ stated in Section 5. The nonzero entries of the sector matrices are $\pm 7, \pm 14, \pm 42$, consistent with the factor of seven proved in Proposition 1.

4.5 Projections between ranks and fortuity

To determine which computed BPS classes can come from larger matrices, we use projection rather than inclusion. In the nontrivial comparisons considered here, the natural inclusion that retains the first N rows and columns, $\mathcal{H}_N \hookrightarrow \mathcal{H}_{N+1}$, is generally not a cochain map. Even on the vacuum, $Q_p^{(N+1)}$ creates terms involving the new index that are absent from the image of $Q_p^{(N)}$. The useful cochain map therefore goes in the opposite direction.

Write

$$W_N := \text{span}\{\Psi_{ij}\}_{1 \leq i, j \leq N}, \quad \mathcal{H}_N = \Lambda^\bullet W_N, \quad \mathcal{B}_R^{(p,N)} := \ker Q_R / \text{im } Q_{R-p}.$$

For $M > N$, let

$$\pi_{M \rightarrow N} : \mathcal{H}_M \rightarrow \mathcal{H}_N$$

be the linear map that deletes every basis monomial containing any generator Ψ_{ij} with $i > N$ or $j > N$. Equivalently, it is the projection onto the subspace generated by monomials using only the first N rows and columns. It preserves fermion number.

Proposition 6 (Projection is a cochain map). *For every odd p and every $M > N$, the projection $\pi_{M \rightarrow N} : \mathcal{H}_M \rightarrow \mathcal{H}_N$ that sends each Ψ_{ij} to itself if $i, j \leq N$ and to zero otherwise, extended multiplicatively to the exterior algebra, satisfies*

$$\pi_{M \rightarrow N} Q_p^{(M)} = Q_p^{(N)} \pi_{M \rightarrow N}.$$

Hence $\pi_{M \rightarrow N}$ induces well defined linear maps on cohomology

$$\Pi_{M \rightarrow N}^{(R)} : \mathcal{B}_R^{(p,M)} \rightarrow \mathcal{B}_R^{(p,N)},$$

and for $L > M > N$ these maps satisfy $\Pi_{L \rightarrow N}^{(R)} = \Pi_{M \rightarrow N}^{(R)} \circ \Pi_{L \rightarrow M}^{(R)}$. The monotone BPS subspace at rank N is

$$\mathcal{B}_{R,\text{mon}}^{(p,N)} := \bigcap_{M > N} \text{Im } \Pi_{M \rightarrow N}^{(R)}.$$

A BPS class is fortuitous in the present family if its coset in $\mathcal{B}_R^{(p,N)} / \mathcal{B}_{R,\text{mon}}^{(p,N)}$ is nonzero. Equivalently, it lies outside $\mathcal{B}_{R,\text{mon}}^{(p,N)}$. Hence, for some $M_0 > N$, it is not in $\text{Im } \Pi_{M_0 \rightarrow N}^{(R)}$. This is the definition used below. Its relation to the covering construction of Ref. [5] is explained in Appendix A.1.

Because the maps compose, the images being intersected can only decrease as M increases. Their intersection is the part of the cohomology at N that lies in the image from every larger rank.

The proof is deferred to Appendix A.1.

Remark 2. In the D1–D5 system [8], the analogous projection does not commute with the supercharge. A preimage of a closed class can then fail to be closed or can become exact at the larger rank. That complication does not arise here.

With respect to the standard embedding $\text{SU}(N) \subset \text{SU}(M)$, the map $\pi_{M \rightarrow N}$ intertwines the $\text{SU}(N)$ actions. Once explicit $\text{SU}(N)$ decompositions are available, the comparison can be made separately for each irreducible representation.

We denote by $\mathcal{B}_{R,\text{fort}}^{(p,N)} := \mathcal{B}_R^{(p,N)} / \mathcal{B}_{R,\text{mon}}^{(p,N)}$ the quotient that measures the independent BPS directions not contained in the monotone subspace.

The reported numerical ranks at $N = 5$ (Table 1) give $h_R^{(5,5)} = 0$ for $R = 0, \dots, 7$. The induced map $\Pi_{5 \rightarrow 4}^{(R)}$ maps the $N = 5$ BPS cohomology in sector R to the $N = 4$ BPS cohomology in the same sector, and the induced maps compose. If the sector at larger N is empty, its image is zero. Conditional on that numerical vanishing, the $N = 4$ BPS classes in that sector are therefore fortuitous under the present definition. Figure 5 illustrates this criterion.

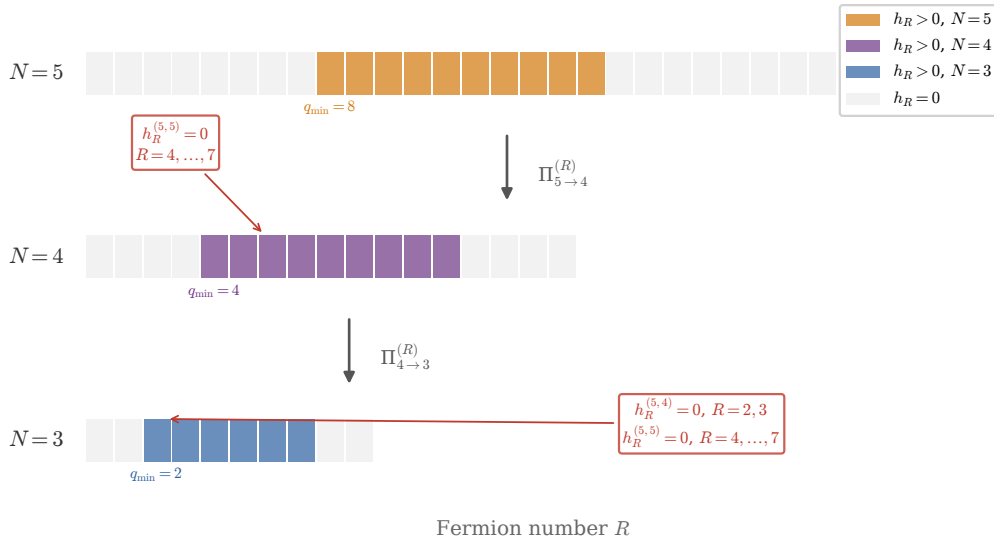


Figure 5: Projection of BPS cohomology from larger to smaller N in the quintic model. Each row shows the sectors at fixed fermion number R . Coloured cells have $h_R > 0$, and the maps $\Pi_{5 \rightarrow 4}^{(R)}$ and $\Pi_{4 \rightarrow 3}^{(R)}$ preserve R . If the BPS space at a larger value of N vanishes, its image at a smaller value of N is zero. Conditional on the displayed numerical vanishings, every nonzero BPS class at $(5, 3)$, and every nonzero BPS class with $R = 4, \dots, 7$ at $(5, 4)$, is fortuitous.

Corollary 7 (Vanishing criterion for fortuity). *If $\mathcal{B}_R^{(p,M_0)} = 0$ for some $M_0 > N$, then $\mathcal{B}_{R,\text{mon}}^{(p,N)} = 0$. Hence every nonzero BPS class in $\mathcal{B}_R^{(p,N)}$ is fortuitous.*

Proof. If $\mathcal{B}_R^{(p,M_0)} = 0$, then $\Pi_{M_0 \rightarrow N}^{(R)} = 0$. By transitivity, every map from a larger rank $L > M_0$ factors through that empty sector:

$$\Pi_{L \rightarrow N}^{(R)} = \Pi_{M_0 \rightarrow N}^{(R)} \circ \Pi_{L \rightarrow M_0}^{(R)} = 0.$$

Hence $\mathcal{B}_{R,\text{mon}}^{(p,N)} = \bigcap_{L > N} \text{Im } \Pi_{L \rightarrow N}^{(R)} = 0$. $\square$

For the first row of Table 2, the required vanishings occur at $N = 4$ for $R = 2, 3$ and at $N = 5$ for $R = 4, \dots, 7$. The second row uses the $N = 5$ quintic vanishings for $R = 4, \dots, 7$. The third uses the computed $p = 7$ vanishings at $N = 5$ for $R \leq 5$. Their numerical status is described in Section 4.1.

p	N	identified	total BPS	fraction	sectors
5	3	440	440	100%	all ($R = 2, \dots, 7$)
5	4	16,125	44,000	37%	$R = 4, \dots, 7$
7	4	4,312	59,136	7%	$R = 3, 4, 5$

Table 2: BPS states identified as fortuitous by projection at fixed fermion number, conditional on the reported numerical vanishings at the larger rank. Particle–hole conjugation gives equal counts in a separate comparison at fixed hole number. Those counts are not added here.

At fixed hole number s , particle–hole conjugation compares the sectors $R_M = M^2 - s$ and $R_N = N^2 - s$ at ranks M and N . This differs from the projection at fixed fermion number, which compares the same value of R at both ranks. A conclusion from the fixed hole comparison therefore does not determine the stable image at fixed fermion number.

Index bound and the numerical difference at $(p, N) = (5, 4)$. For $p = 5$, $N = 4$, the Euler characteristics in the five residue classes are obtained from (20):

$$(I_0, I_1, I_2, I_3, I_4) = (3625, 3625, -9500, 11750, -9500). \quad (54)$$

Therefore the exact lower bound from the indices is

$$\sum_{a=0}^4 |I_a| = 38,000. \quad (55)$$

Comparing with the computed BPS count (42) gives the difference

$$44,000 - 38,000 = 6,000. \quad (56)$$

The difference is not distributed uniformly across residue classes (Figure 6(b)). Writing $Z_a = \sum_{R \equiv a \pmod{5}} h_R$, one finds

residue a	$ I_a $	Z_a	difference
0	3625	6375	2750
1	3625	6375	2750
2	9500	9750	250
3	11750	11750	0
4	9500	9750	250

In residue 3 the BPS count equals the index bound, while residues 0 and 1 give the largest differences. This decomposition compares, residue class by residue class, the computed BPS count with the lower bound supplied by the protected index. It does not select individual BPS states.

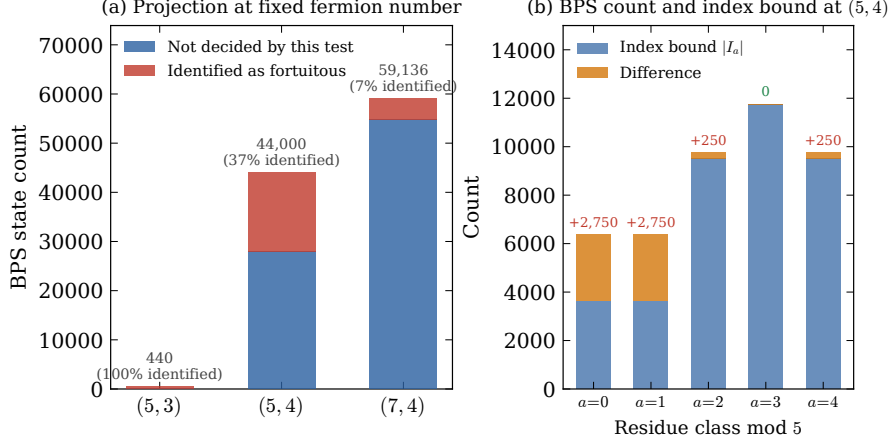


Figure 6: Projection test and index comparison. Left panel: red denotes states conditionally identified as fortuitous by the projection test, using the reported numerical vanishings at larger N . Blue denotes states whose status is not determined by this test. Percentages refer only to this comparison. Right panel: decomposition by residue class at $(5, 4)$, showing the lower bound $|I_a|$ (blue) and the difference from that bound (orange) in each class $a \bmod 5$. The BPS count in residue $a = 3$ equals the bound.

5 Factorization and map ranks

The BPS generating functions in Section 4 display a common factorization pattern. We now state that pattern precisely and relate it to the ranks of the maps Q_R . Table 3 collects the main numerical results for the four nontrivial cases with data in every sector.

(p, N)	2^{N^2}	$\sum h_R$	index bound	difference	$q_{\min}$	$p^{b_{p,N}}$	$\deg T$	$T(1)$	$\#\{E/p^2\}$
(5, 3)	512	440	400	40	2	5^1	2	11	3
(5, 4)	65,536	44,000	38,000	6,000	4	5^3	4	22	9
(7, 4)	65,536	59,136	55,468	3,668	3	7^1	6	528	8
(5, 5)	33,554,432	15,480,000	11,781,250	3,698,750	8	5^4	4	774	—

Table 3: Summary of the numerical results at fixed N . For each computed case, the columns give the dimension of the Hilbert space, the total BPS count, the lower bound from the Witten indices and the difference between them, the minimum BPS charge $q_{\min}$, the power $p^{b_{p,N}}$, the degree and evaluation $T(1)$ of the final quotient, and the number of distinct rescaled energy levels E/p^2 .

Whenever $(1+x)^N$ divides $Z_{\text{BPS}}^{(p,N)}(x)$, set

$$P_N^{(p)}(x) := x^{-q_{\min}}(1+x)^{-N} Z_{\text{BPS}}^{(p,N)}(x). \quad (57)$$

Here $q_{\min}$ is the minimum BPS charge. If $P_N^{(p)}$ has integer coefficients, let $b_{p,N}$ be the largest integer for which $p^{b_{p,N}}$ divides all of them, and define

$$T_N^{(p)}(x) := p^{-b_{p,N}} P_N^{(p)}(x).$$

Corollary 3 accounts for two factors of $1+x$ when N is even and three when N is odd. Thus the observed factor $(1+x)^N$ contains $N-2$ further factors when N is even and $N-3$ when N is odd. In particular, the factor $(1+x)^3$ at $(p, N) = (5, 3)$ is entirely structural. The additional divisibility first appears in the present data at $N = 4$.

Particle–hole conjugation gives $Z_{\text{BPS}}^{(p,N)}(x) = x^{N^2} Z_{\text{BPS}}^{(p,N)}(x^{-1})$. Therefore, whenever $x^{q_{\min}}(1+x)^N$ divides it, the quotient is automatically palindromic.

Conjecture 1 (Additional factors). *For odd $p \geq 5$ with $p \leq 2N - 1$,*

$$(1+x)^N \mid Z_{\text{BPS}}^{(p,N)}(x) \quad \text{in } \mathbb{Z}[x]. \quad (58)$$

After the minimum charge and this factor are removed, the quotient has nonnegative coefficients and a common positive factor of p :

$$P_N^{(p)}(x) \in p\mathbb{Z}_{\geq 0}[x]. \quad (59)$$

Equivalently, $b_{p,N} \geq 1$ and

$$Z_{\text{BPS}}^{(p,N)}(x) = p^{b_{p,N}} x^{q_{\min}} (1+x)^N T_N^{(p)}(x), \quad (60)$$

where $T_N^{(p)}(x)$ has nonnegative integer coefficients and its coefficients do not all have a further factor of p . Its palindromicity follows from particle–hole conjugation.

For $(p, N) = (5, 3), (5, 4), (7, 4)$, the computed parameters satisfy the simple formulas

$$q_{\min} = \binom{N-m}{2} + m, \quad b_{p,N} = \binom{N-m}{2}, \quad m = \frac{p-3}{2}. \quad (61)$$

These formulas describe the three numerical data sets with $N \leq 4$. They are not claims for arbitrary N . These patterns do not extend to $N = 5$: the computations give $q_{\min} = 8$ at $(5, 5)$ and $q_{\min} = 6$ at $(7, 5)$, each one unit above the $N \leq 4$ value. They also give $b_{5,5} = 4$, rather than the extrapolated value 6. No explicit formula for $q_{\min}$ or $b_{p,N}$ valid for arbitrary N is currently known. The numerical polynomial at $(5, 5)$ still has the form (60). The ten reported ranks below the middle and rank symmetry leave only the middle rank undetermined. The exact relation $(1+x) \mid \mathcal{R}_{5,5}(x)$ fixes that rank, and the direct numerical calculation of h_{10} agrees with (52). Table 4 compares the two calculations at $N = 5$.

p	N	$q_{\min}$	$p^{b_{p,N}}$	$\deg T$	total BPS	sectors computed
5	5	8	5^4	4	15,480,000	all
7	5	6	—	—	—	$R \leq 6$

Table 4: Numerical results at $N = 5$. Both values of $q_{\min}$ are one above the formulas inferred from $N \leq 4$ in (61). The dashes in the $p = 7$ row require computations beyond $R = 6$.

All four numerical polynomials have the factor $(1+x)^N$, a common positive power of p , and a final quotient with nonnegative coefficients. Palindromicity of that quotient follows from particle–hole conjugation once the divisibility is known. The sequence $b_{5,N} = 1, 3, 4$ for $N = 3, 4, 5$ has no known explicit formula.

After the factors proved in Corollary 3 are removed, the conjectural content is the remaining divisibility needed to reach $(1+x)^N$, the positive common factor of p , and nonnegativity of the final quotient.

Theorem 3 (Identity relating the BPS polynomial to the map ranks). *Define the polynomial*

$$\mathcal{R}_{p,N}(x) := \sum_{R \geq 0} r_R x^R, \quad r_R := \text{rank } Q_R = \text{rank } \tilde{Q}_{p,R}. \quad (62)$$

Then

$$Z_{\text{BPS}}^{(p,N)}(x) = (1+x)^{N^2} - (1+x^p) \mathcal{R}_{p,N}(x), \quad (63)$$

and consequently, for every $1 \leq k \leq N^2$,

$$(1+x)^k \mid Z_{\text{BPS}}^{(p,N)}(x) \iff (1+x)^{k-1} \mid \mathcal{R}_{p,N}(x). \quad (64)$$

In particular, $(1+x)^N \mid Z_{\text{BPS}}^{(p,N)}(x)$ if and only if $(1+x)^{N-1} \mid \mathcal{R}_{p,N}$. Thus the factor $(1+x)^N$ is equivalent to a divisibility property of the ranks r_R . Any explanation of the factor $(1+x)^N$ should therefore also account for the corresponding divisibility of $\mathcal{R}_{p,N}(x)$.

Proof. Equation (63) follows from (13) by summing $h_R = \dim V_R - r_R - r_{R-p}$ against x^R and using $\sum_R \binom{N^2}{R} x^R = (1+x)^{N^2}$. For (64), since $1 \leq k \leq N^2$, the term $(1+x)^{N^2}$ in (63) is divisible by $(1+x)^k$. Hence

$$(1+x)^k \mid Z_{\text{BPS}}^{(p,N)}(x) \iff (1+x)^k \mid (1+x^p) \mathcal{R}_{p,N}(x).$$

For odd p ,

$$1+x^p = (1+x)(1-x+x^2-\dots+x^{p-1}),$$

and the second factor takes the nonzero value p at $x = -1$. It is therefore coprime to $1+x$ over $\mathbb{Q}[x]$, so

$$(1+x)^k \mid (1+x^p) \mathcal{R}_{p,N}(x) \iff (1+x)^{k-1} \mid \mathcal{R}_{p,N}(x).$$

Because powers of $1+x$ are monic, the same divisibility statements hold over $\mathbb{Z}[x]$. This proves (64). $\square$

The reported rank data exhibit this stronger divisibility. The polynomials $\mathcal{R}_{p,N}$ factor as

$$\mathcal{R}_{5,3}(x) = (1+x)^2(1+7x+x^2), \quad (65)$$

$$\mathcal{R}_{5,4}(x) = (1+x)^3(1+13x+78x^2+286x^3+590x^4+286x^5+78x^6+13x^7+x^8), \quad (66)$$

$$\mathcal{R}_{7,4}(x) = (1+x)^3(1+13x+78x^2+216x^3+78x^4+13x^5+x^6), \quad (67)$$

The parenthetical factor in each line has positive integer coefficients. Theorem 1 proves the palindromicity of $\mathcal{R}_{p,N}$. Since $\mathcal{R}_{p,N}$ and $(1+x)^{N-1}$ are palindromic, the quotient is automatically palindromic whenever the divisibility holds. The exterior algebra of creation operators acts by maps of the complex. Every creation operator $c = \Psi_{ij}$ supercommutes with Q_p :

$$c Q_p = -Q_p c, \quad (68)$$

since both are odd and built purely from creators. Hence left multiplication by c sends $\text{im } Q_R$ into $\text{im } Q_{R+1}$ and $\text{ker } Q_R$ into $\text{ker } Q_{R+1}$, and therefore acts on cohomology. Section 6 considers whether the $N-1$ traceless diagonal creation modes can explain the remaining factors of $1+x$.

Proposition 8 (Determination of the middle rank at $(5,5)$). *Conditional on the ten reported numerical ranks $r_0, \dots, r_9$, the proved rank symmetry and the exact relation $(1+x) \mid \mathcal{R}_{5,5}(x)$ determine the middle rank r_{10} . For the reconstructed numerical polynomial $\mathcal{R}_{5,5}$, the reported ranks satisfy the additional relation that gives $(1+x)^4 \mid \mathcal{R}_{5,5}$, while $(1+x)^5 \nmid \mathcal{R}_{5,5}$.*

Proof. Let $P(x) = x^{2d}P(1/x)$ be palindromic of even degree $2d$. If $P(-1) = 0$, then $P/(1+x)$ is palindromic of odd degree and therefore also vanishes at -1 . Thus a first root at -1 has multiplicity at least two.

At $(p, N) = (5, 5)$, the polynomial $\mathcal{R}_{5,5}$ has degree 20. Corollary 3 gives the exact condition $\mathcal{R}_{5,5}(-1) = 0$. Rank symmetry expresses every coefficient except r_{10} in terms of $r_0, \dots, r_9$. The exact condition is therefore one linear equation for r_{10} , and it gives

$$r_{10} = 2,047,506.$$

The reconstructed value agrees with the direct numerical calculation at $R = 10$ reported in Section 4.4. Palindromicity then supplies the second factor of $1+x$.

The quotient $\mathcal{R}_{5,5}/(1+x)^2$ is palindromic of even degree 18. The reported ranks satisfy the additional condition that this quotient vanishes at -1 , and palindromicity supplies the fourth factor. The next obstruction is

$$\left. \frac{\mathcal{R}_{5,5}(x)}{(1+x)^4} \right|_{x=-1} = 13,750 = 2 \cdot 5^4 \cdot 11 \neq 0.$$

□

For the reconstructed numerical polynomial $\mathcal{R}_{5,5}$,

$$\mathcal{R}'_{5,5}(-1) = \mathcal{R}''_{5,5}(-1) = \mathcal{R}'''_{5,5}(-1) = 0, \quad \mathcal{R}^{(4)}_{5,5}(-1) = 330,000 \neq 0.$$

Theorem 3 gives the factor $(1+x)^5$ in the reconstructed numerical BPS polynomial (52), which can be written as

$$Z_{\text{BPS}}^{(5,5)}(x) = 5^4 x^8 (1+x)^5 (54 + 221x + 224x^2 + 221x^3 + 54x^4). \quad (69)$$

Thus the reconstruction uses the ten numerical ranks below the middle, the proved symmetry of the ranks, and the exact relation $(1+x) \mid \mathcal{R}_{5,5}(x)$. It does not assume the further factors observed in the data. The direct calculation at $R = 10$ supplies an independent numerical check.

Remark 3 (Relation to the cubic model). Reference [3] found the exact total generating function

$$Z_{\text{BPS}}^{(3,N)}(x) = 3^{\binom{N}{2}} x^{\binom{N}{2}} (1+x)^N. \quad (70)$$

Thus the factor $(1+x)^N$ and a positive power of the supercharge degree already occur for $p = 3$, with $b_{3,N} = q_{\min} = \binom{N}{2}$ and $T = 1$. For $p \geq 5$, the question is whether this structure persists when the Hamiltonian is no longer determined in general by the quadratic Casimir.

6 Discussion and open questions

The numerical generating polynomials in Eqs. (44), (45), (49), and (52) are the main computational results. All four contain $(1+x)^N$. The data at $N = 5$ also show that the formulas for $q_{\min}$ and $b_{p,N}$ inferred from $N \leq 4$ do not continue unchanged. The factorization conjecture remains consistent with the four polynomials, but its parameters are not known in general.

Corollary 3 proves divisibility by $(1+x)^2$ for every $N \geq 2$, and by $(1+x)^3$ when N is odd. Theorem 1 gives the rank symmetry for odd $p \leq 2N - 1$. Theorem 3 expresses the remaining divisibility as a condition on $\mathcal{R}_{p,N}$, but these results do not prove $(1+x)^N$ for arbitrary N .

The Hamiltonian analysis addresses a different question. In the numerical (5,4), $R = 4$ example, equivalent $\text{SU}(4)$ summands occur at different energies. The exact normal ordering identities show that the combination $-K_3 + K_4$ in $\tilde{H} = H/25$ produces this splitting. This

result concerns the energy spectrum, whereas $(1+x)^N$ concerns multiplicities at zero energy. The formulas for the operators K_k and their commutators do not explain that factor.

The projections between matrix sizes provide a separate comparison of the cohomology. Conditional on the reported numerical vanishings at larger N , Table 2 lists the sectors and numbers of BPS classes identified as fortuitous at fixed fermion number. The stable images in the remaining nonzero sectors are not yet determined.

6.1 Comparison with supersymmetric SYK

Here the support means the fermion numbers for which $h_R \neq 0$. It contains 9 of 17 sectors at $(p, N) = (5, 4)$, 11 of 17 at $(7, 4)$, and 10 of 26 at $(5, 5)$.

For comparison, consider $\mathcal{N} = 2$ SYK with the same degree $\hat{q}$ of the supercharge and the same number $N_f = N^2 = 16$ of complex fermions [6, 39]. The examples obtained by exact diagonalization in those references have BPS states in $\hat{q}$ sectors, and their total count equals the lower bound from the Witten indices. The corresponding values are shown below and in Figure 7.

Model	Z_{BPS}	Index bound	Difference	Sectors
SYK $\hat{q} = 3, N_f = 16$	8,748	8,748	0	3
SYK $\hat{q} = 5, N_f = 16$	38,000	38,000	0	5
SYK $\hat{q} = 7, N_f = 16$	55,468	55,468	0	7
Matrix (5, 4)	44,000	38,000	6,000	9
Matrix (7, 4)	59,136	55,468	3,668	11

The comparison uses the lower bound from the Witten indices in the $\hat{q}$ residue classes. For the two matrix models, the differences are 6,000 at $(5, 4)$ and 3,668 at $(7, 4)$. For $\hat{q} = 7$, the two central SYK multiplicities, $h_7 = 11,319$ and $h_8 = 12,838$ [6], coincide with the numerical matrix model values at $(7, 4)$. Differences occur away from the center and in the additional outer sectors of the matrix model.

The distribution of supersymmetric ground states across charge sectors has also been analysed in the double scaling limit [44]. It shapes the phase structure at nonzero background charge [45]. Its relation to the Schwarzian description of black holes close to the BPS bound is developed in Refs. [6, 46, 47].

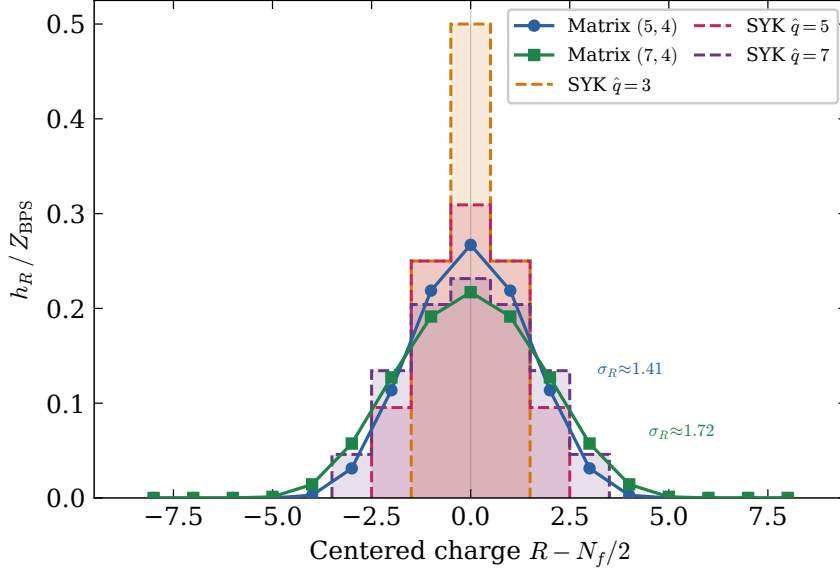


Figure 7: Charge support: matrix model vs. $\mathcal{N} = 2$ SYK. Normalized numerical BPS distributions h_R/Z_{BPS} for the matrix models (5, 4) and (7, 4) (solid lines, $N^2 = 16$) and the $\mathcal{N} = 2$ SYK examples at $\hat{q} = 3, 5, 7$ (shaded bars, $N_f = 16$). Every distribution shown concerns a system with 16 complex fermions. The labels give the standard deviation from half filling, $\sigma_R = [\sum_R (R - 8)^2 h_R/Z_{\text{BPS}}]^{1/2}$ for each matrix model distribution. In each computed case, SYK has BPS states in $\hat{q}$ sectors and its BPS count equals the index bound. The matrix model has nonzero multiplicities in more sectors and lies strictly above the index bound.

Tensor models without disorder can reproduce the melonic large N limit of SYK [48, 49].

Reference [20] also studies a model without disorder in which $2j + 1$ complex fermions transform in the spin j representation of $\text{SU}(2)$. For observables that are singlets under $\text{SU}(2)$, its expansion at large j is melonic. At small charge and low energy, the resulting equations agree with those of $\mathcal{N} = 2$ SYK and contain the corresponding Schwarzian mode. The present model instead has an exact $\text{SU}(N)$ symmetry that acts by conjugation on the matrix fermions. No melonic limit is constructed here. Reference [6] relates the concentration of BPS states in a narrow range of R to fortuity in supersymmetric SYK models. It also formulates a conjecture concerning quantum chaos and the supercharge. The comparison in Figure 7 does not by itself determine fortuity under the projections at fixed R used here or provide a test of chaos for the present matrix model.

Another comparison is the $\mathcal{N} = 2$ gauged quantum mechanics obtained by reducing a Chern–Simons matter theory in three dimensions on S^2 [50]. It reproduces the entropy of BPS black holes in AdS_4 and has statistically distributed couplings, but it arises from string theory rather than from a supercharge given by a single trace.

6.2 Bound at large N

Across the three values of N studied numerically, the proportion of BPS states decreases while the total BPS count increases, as shown in Table 5.

N	Z_{BPS}	2^{N^2}	BPS/total	$\frac{1}{N^2} \log Z_{\text{BPS}}$
3	440	512	85.9%	0.676
4	44,000	65,536	67.1%	0.668
5	15,480,000	33,554,432	46.1%	0.662

Table 5: Reported numerical quintic BPS data for $N = 3, 4, 5$. The BPS fraction decreases across the three displayed values, while the normalized entropy remains near 0.66. Whether either trend persists is open.

The three values do not determine whether $N^{-2} \log Z_{\text{BPS}}$ has a limit. The Witten indices give the following result for every fixed p .

Theorem 4 (Bound from the Witten indices). *Let $p \geq 3$ be fixed and odd. The indices in (20) obey*

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \log \sum_{a=0}^{p-1} |I_a| = \log(2 \cos(\pi/(2p))). \quad (71)$$

Consequently,

$$\liminf_{N \rightarrow \infty} \frac{1}{N^2} \log Z_{\text{BPS}}^{(p,N)} \geq \log(2 \cos(\pi/(2p))). \quad (72)$$

Moreover, for any sequence of matrix sizes $N_j \rightarrow \infty$ along which $N_j^{-2} \log Z_{\text{BPS}}^{(p,N_j)}$ converges, its limit lies in

$$[\log(2 \cos(\pi/(2p))), \log 2]. \quad (73)$$

Proof. Put $n = N^2$ and $b_\ell = (1 - \omega^\ell)^n$. Equation (20) is the inverse discrete Fourier transform of (b_ℓ) . Expanding its squared ℓ^2 -norm and using the root-of-unity orthogonality relation $\sum_{a=0}^{p-1} \omega^{a(\ell-\ell')} = p \delta_{\ell,\ell'}$ gives

$$\sum_{a=0}^{p-1} |I_a(n)|^2 = \frac{1}{p} \sum_{\ell=0}^{p-1} |1 - \omega^\ell|^{2n}. \quad (74)$$

For $\ell = 1, \dots, p-1$, $|1 - \omega^\ell| = 2 \sin(\pi\ell/p)$. Since p is odd, the maximum $\lambda = 2 \cos(\pi/(2p))$ is attained exactly at $\ell = (p \pm 1)/2$. Therefore, for $I(n) = (I_0(n), \dots, I_{p-1}(n))$,

$$\sqrt{\frac{2}{p}} \lambda^n \leq \|I(n)\|_2 \leq \|I(n)\|_1 \leq \sqrt{p} \|I(n)\|_2 \leq \sqrt{p-1} \lambda^n.$$

Taking logarithms, dividing by $n = N^2$, and letting $N \rightarrow \infty$ proves (71). The signed cohomology identity above gives $Z_{\text{BPS}} \geq \sum_a |I_a|$, and hence (72). Finally, $Z_{\text{BPS}} \leq 2^{N^2}$, which gives the upper bound in (73). $\square$

Numerically, the bound is 0.549 at $p = 3$, 0.643 at $p = 5$, and 0.668 at $p = 7$, approaching $\log 2 \approx 0.693$ as $p \rightarrow \infty$.

The observed quintic sequence 0.676, 0.668, 0.662 for $N = 3, 4, 5$ (Table 5) lies above 0.643, with a decreasing gap. This is consistent with $\frac{1}{N^2} \log Z_{\text{BPS}}^{(5,N)}$ approaching the rate of the index bound from above. The three available values do not distinguish whether the sequence has the lower rate in (72) or a larger one.

6.3 Open questions

The first open question is whether $(1+x)^N$ divides the BPS polynomial for every odd $p \geq 5$ with $p \leq 2N - 1$. The cubic case is known exactly. Corollary 3 proves two factors for even N and three factors for odd N . The additional factors seen in the data remain unexplained.

Let

$$\mathcal{B}^\bullet := \bigoplus_R \mathcal{B}_R^{(p,N)},$$

and let $\mathcal{B}_0^\bullet$ denote the cohomology of the traceless complex. Theorem 2 and the Künneth formula give an isomorphism of graded vector spaces

$$\mathcal{B}^\bullet \cong \Lambda^\bullet(\mathbb{C}\chi) \otimes \mathcal{B}_0^\bullet.$$

Each traceless linear combination of the diagonal creation operators anticommutes with Q_p and therefore induces a map of degree one on $\mathcal{B}_0^\bullet$. The traceless diagonal combinations form a vector space of dimension $N - 1$. Their induced maps anticommute and square to zero, so they make $\mathcal{B}_0^\bullet$ a graded module over $\Lambda^\bullet(\mathbb{C}^{N-1})$.

A sufficient explanation for the remaining factors of $1 + x$ would be the existence of a graded vector space $\mathcal{M}^{(p,N)}$ and an isomorphism of graded modules

$$\mathcal{B}_0^\bullet \cong \Lambda^\bullet(\mathbb{C}^{N-1}) \otimes \mathcal{M}^{(p,N)}.$$

Write $\text{Hilb } \mathcal{M}^{(p,N)}(x)$ for the Hilbert series of $\mathcal{M}^{(p,N)}$. Taking Hilbert series would then give

$$Z_{\text{BPS}}^{(p,N)}(x) = (1 + x)^N \text{Hilb } \mathcal{M}^{(p,N)}(x).$$

The quotient by $(1 + x)^N$ would therefore have nonnegative integer coefficients. If the full factorization in (60) holds, then

$$\text{Hilb } \mathcal{M}^{(p,N)}(x) = p^{b_{p,N}} x^{q_{\min}} T_N^{(p)}(x).$$

The freeness statement would not by itself explain the common power of p . Such freeness is not implied by divisibility of the Hilbert series and is not proved here. For broader context, Refs. [51, 52] develop Hilbert series for Coulomb branches of three-dimensional $\mathcal{N} = 4$ theories.

The second question concerns the exponent $b_{p,N}$. The sequence $b_{5,N} = 1, 3, 4$ for $N = 3, 4, 5$ has no known explicit formula. At $N = 5$, the reported 5^4 divisibility of h_R does not arise from 5^4 divisibility of $\dim \ker Q_R$ and $\text{rank } Q_{R-5}$ separately. In the first two surviving sectors $R = 8, 9$, the nullities and incoming ranks are divisible by 5^2 , while at $R = 10$ they are not. Nevertheless, the differences h_R are all divisible by 5^4 . For the reported numerical ranks, the relation is

$$\dim \ker Q_R \equiv \text{rank } Q_{R-5} \pmod{5^4},$$

although neither term on the two sides is always divisible by 5^4 . Explaining this congruence may help determine $b_{p,N}$.

The third question is the behaviour as N grows. At fixed p , Theorem 4 gives the interval (73), but the available values of N do not determine the limiting rate. A different problem is the joint limit along odd integers $p = p(N)$ that satisfy $p(N) \leq 2N - 1$ and $p(N)/N \rightarrow \alpha \in (0, 2]$. The present calculation does not determine the asymptotic behaviour of $N^{-2} \log Z_{\text{BPS}}^{(p(N),N)}(1)$ in this regime.

A separate question is whether the model admits, for every $N \geq 3$, a family of pairwise commuting conserved operators that are invariant under $\text{SU}(N)$, go beyond functions of H , and resolve equivalent irreducible summands. The nonzero commutators among the displayed K_k show that these operators cannot form such a family in the stated $N = 4$ sectors. They do not prove chaos or a failure of integrability, and they do not exclude a different family.

Finally, the stable images at fixed fermion number remain undetermined for $R = 8, \dots, 12$ at $(p, N) = (5, 4)$ and for $R = 6, \dots, 13$ at $(7, 4)$. No stable image of a nonzero BPS sector at $N = 5$ is determined by the present data, because that would require cohomology at larger N . At rank N , particle-hole conjugation sends sector R to sector $N^2 - R$. Across different values of N , the latter sectors have a fixed number of holes rather than a fixed fermion number, so conjugation does not determine the stable images at fixed R .

Acknowledgments

The author thanks Hank Chen, Yiming Chen, Leonardo Santilli, Jan Troost and Alberto Zaffaroni for helpful comments and correspondence.

Appendix A Deferred proofs and algebraic background

This appendix supplies the proofs deferred from the main text and summarizes the facts about cohomology of Lie algebras used in the model.

A.1 Projections between ranks and fortuity

Proof of Proposition 6. Since $\pi_{M \rightarrow N}$ is defined as an algebra homomorphism on $\mathcal{H}_M$ (sending each Ψ_{ij} to itself if $i, j \leq N$ and to zero otherwise, and extending multiplicatively), write $q_p^{(N)} := \sum_{i_1, \dots, i_p=1}^N \Psi_{i_1 i_2} \cdots \Psi_{i_{p-1} i_p}$ so that $Q_p^{(N)}(v) = q_p^{(N)} \wedge v$. Then $\pi_{M \rightarrow N}(q_p^{(M)}) = q_p^{(N)}$, since the projection sets to zero every term in $q_p^{(M)}$ involving an index $> N$. Therefore for all $v \in \mathcal{H}_M$:

$$\begin{aligned} \pi_{M \rightarrow N}(Q_p^{(M)}v) &= \pi_{M \rightarrow N}(q_p^{(M)} \wedge v) \\ &= \pi_{M \rightarrow N}(q_p^{(M)}) \wedge \pi_{M \rightarrow N}(v) \\ &= q_p^{(N)} \wedge \pi_{M \rightarrow N}(v) = Q_p^{(N)}\pi_{M \rightarrow N}(v), \end{aligned}$$

proving the intertwining identity for all $M > N$. The intertwining identity shows that $\pi_{M \rightarrow N}$ sends $\ker Q_p^{(M)}$ into $\ker Q_p^{(N)}$ and $\text{im } Q_p^{(M)}$ into $\text{im } Q_p^{(N)}$, so the induced maps on cohomology are well defined. Transitivity follows from $\pi_{L \rightarrow N} = \pi_{M \rightarrow N} \circ \pi_{L \rightarrow M}$. The images $\text{Im } \Pi_{M \rightarrow N}^{(R)}$ form a descending chain in the space $\mathcal{B}_R^{(p, N)}$, which has finite dimension. For $L > M > N$, transitivity gives $\text{Im } \Pi_{L \rightarrow N}^{(R)} \subseteq \text{Im } \Pi_{M \rightarrow N}^{(R)}$. The finite dimension ensures that the chain stabilizes, so $\mathcal{B}_{R, \text{mon}}^{(p, N)}$ equals the eventual stable image. $\square$

Relation to the covering definition. Reference [5] defines the monotone subspace as the image in cohomology of a map from one covering complex. The present paper instead uses the eventual image of the maps $\Pi_{M \rightarrow N}^{(R)}$ as M increases. Proposition 6 shows that these images are well defined in cohomology and stabilize at each fixed fermion number. This is analogous to the image used in Ref. [5], but no equivalence of the two constructions is assumed. In the D1–D5 system, the analogous projection need not commute with the supercharge [8]. Here commutation follows directly because $Q_p = \text{Tr}(\Psi^p)$ contains only creation operators.

A.2 Hodge duality at $(p, N) = (5, 3)$

Proof of Proposition 5. Throughout this proof a superscript records the model to which an object of Section 3 belongs. The map $\mathcal{A}_k^{(p)}$ is defined in (24) using Ω_p , and $K_k^{(p, N)}$ is the operator in (27) for the model with parameters (p, N) . Here $N = 3$ throughout, and K_k means $K_k^{(5, 3)}$.

Work first on the traceless Fock space $\Lambda^\bullet W_0$, where $W_0 \simeq \mathfrak{sl}_3$. Fix an orthonormal basis $\{T_a\}_{a=1}^8$ of traceless Hermitian matrices and let $W_{0, \mathbb{R}}$ be its real span. Thus $W_0 = W_{0, \mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, and the adjoint action of $\text{SU}(3)$ on $W_{0, \mathbb{R}}$ lies in $SO(8)$. We use the complex linear Hodge star obtained by extending the real Euclidean Hodge star on $W_{0, \mathbb{R}}$. In the basis T_a it is a signed permutation of basis wedges, so it is unitary for the induced Hermitian inner product and commutes with the $\text{SU}(3)$ action. The relevant dimension equality is

$$\dim_{\mathbb{C}} W_0 = N^2 - 1 = 8 = 3 + 5,$$

special to $N = 3$, which makes $*$ carry $\Lambda^3 W_0^*$ isomorphically onto $\Lambda^5 W_0^*$. The spaces of invariant forms of degrees three and five on W_0 both have dimension one [31, 32]. Therefore, by equivariance,

$$\Omega_5 = c * \Omega_3 \quad (75)$$

for some nonzero complex scalar c . Only the modulus of c is needed below. In the convention fixed by $\tilde{Q}_p = p^{-1} \text{Tr}(A^p)$, (30) and (37) give $\|\Omega_5\|^2 = 24$ and $\|\Omega_3\|^2 = 8$. Since $*$ is unitary, (75) forces $|c|^2 = 3$. This is the $\text{SU}(3)$ duality between the primitive cubic and quintic cocycles [34, 42].

Set $q = \tilde{Q}_3$, so that q acts on $\Lambda^\bullet W_0$ as exterior multiplication by Ω_3 , and put $B = q^\dagger q$. We claim that, on $\Lambda^k W_0$ for $0 \leq k \leq 5$,

$$(\mathcal{A}_k^{(5)})^\dagger \mathcal{A}_k^{(5)} = 3 B|_{\Lambda^k W_0}. \quad (76)$$

We use the standard relation between contraction of a Hodge dual and exterior multiplication. Write $\flat$ for the complex linear isomorphism $\Lambda^k W_0 \rightarrow \Lambda^k W_0^*$ obtained from the complex bilinear extension of the real Euclidean form for which the T_a are orthonormal. Then, for $\alpha \in \Lambda^k W_0$ and $\omega \in \Lambda^m W_0^*$,

$$\iota_\alpha(*\omega) = \varepsilon *(\omega \wedge \alpha^\flat), \quad (77)$$

where $\varepsilon \in \{\pm 1\}$ depends only on k, m and $\dim W_0$, not on α or ω . For $k = 1$ this is the usual relation $\iota_v(*\omega) = *(\omega \wedge v^\flat)$ [53]. The general case follows by iterating it, each step contributing a sign that depends only on the degrees involved. We suppress $\flat$ from here on and use this complex linear identification of W_0 with W_0^* . Applying (77) with $\omega = \Omega_3$ and using (75), we get, for all $\alpha, \beta \in \Lambda^k W_0$,

$$\langle \mathcal{A}_k^{(5)} \alpha, \mathcal{A}_k^{(5)} \beta \rangle = |c|^2 \langle *(\Omega_3 \wedge \alpha), *(\Omega_3 \wedge \beta) \rangle = 3 \langle \Omega_3 \wedge \alpha, \Omega_3 \wedge \beta \rangle = 3 \langle q^\dagger q \alpha, \beta \rangle.$$

The sign ε cancels because it occurs in both arguments of the sesquilinear form. The middle equality uses the unitarity of $*$. This is (76). This identity, including the factor 3, is independently verified by exact arithmetic for every k in the ancillary program `verify_normal_ordering.py`.

After normal ordering, let $B_{[j]}$ denote the part of B containing j creation and j annihilation operators. Then

$$B = \sum_{j=0}^3 B_{[j]}.$$

Its components follow from the cubic identities (37) and from the cancellation of the term with three creation and three annihilation operators in $\{q, q^\dagger\}$:

$$B_{[0]} = 8, \quad B_{[1]} = -3\hat{n}, \quad B_{[2]} = 3\hat{n} - \hat{C}_2, \quad B_{[3]} = -qq^\dagger. \quad (78)$$

It remains to lift (76) from the sector with k particles to the full Fock space. By (27), K_k preserves fermion number and its matrix on the sector with k particles is $(\mathcal{A}_k^{(5)})^\dagger \mathcal{A}_k^{(5)}$. For $|I| = |J| = k$, let $|I\rangle = a_I^\dagger |0\rangle$ and define $B_{I,J} := \langle I|B|J\rangle$. By (76), the (I, J) entry of this matrix is $3B_{I,J}$. More explicitly, consider a term $a_{I'}^\dagger a_{J'}$ in $B_{[j]}$ written in normal order, with $|I'| = |J'| = j$. Its contribution to the k -particle matrix is obtained by adjoining the same $(k-j)$ -element set T , disjoint from $I' \cup J'$, to the input and output index sets. The signs required to order $I' \cup T$ and $J' \cup T$ occur once in the matrix element and once in the corresponding operator in normal order, so they cancel. Let $|O\rangle$ be a basis state with r occupied modes. If $a_{I'}^\dagger a_{J'} |O\rangle \neq 0$, then $J' \subset O$ and $I' \cap (O \setminus J') = \emptyset$. The allowed sets T are therefore exactly the $(k-j)$ -element subsets of $O \setminus J'$, of which there are $\binom{r-j}{k-j}$. If the original term annihilates

$|O\rangle$, every term obtained by adjoining T also vanishes. The lift of $B_{[j]}$ is therefore $\binom{\hat{n}-j}{k-j}B_{[j]}$. Consequently

$$K_k^{(5,3)} = 3 \sum_{j=0}^{\min(3,k)} \binom{\hat{n}-j}{k-j} B_{[j]}, \quad 0 \leq k \leq 4, \quad (79)$$

as an identity of operators on each sector with fixed $\hat{n}$. This identity is also checked exactly in the ancillary program. The four operators in (78) commute: the first three are functions of $\hat{n}$ and $\hat{C}_2$, while $qq^\dagger$ preserves $\hat{n}$ and is $SU(3)$ -invariant, hence commutes with both. Since $\hat{n}$ commutes with all of them, every $K_k^{(5,3)}$ in (79) is a polynomial in the four commuting operators $\hat{n}$, $\hat{C}_2$, $qq^\dagger$ and the identity. Equation (79) therefore proves that the five operators $K_0, \dots, K_4$ commute pairwise. Finally, on the full Fock space they act as $K_k|_{\Lambda^\bullet W_0} \otimes \mathbf{1}_{\Lambda^\bullet(\chi)}$, so the result extends to the trace factor. $\square$

A.3 Cohomological background

Relation to standard cohomological complexes. The BPS space is the cohomology of

$$(\Lambda^\bullet(\mathfrak{gl}_N), Q_p \wedge).$$

The differential has degree p , so fixing fermion number modulo p gives p independent complexes. After the exact $U(1)$ decoupling of Section 2.4, the complex is

$$\Lambda^\bullet(\chi) \otimes (\Lambda^\bullet(\mathfrak{sl}_N), Q_p \wedge).$$

This is not the Chevalley–Eilenberg complex [31]. That differential includes contraction. For odd $p \geq 3$, the differential here is exterior multiplication by a primitive cocycle of degree p when $p \leq 2N - 1$, and it vanishes when $p > 2N - 1$. In the Aomoto complex, exterior multiplication by a class of degree one acts on an Orlik–Solomon algebra [36, 37]. Here the underlying algebra is the full exterior algebra and the multiplying class has degree p .

What is classical is the existence of the primitive generators and the structure of the invariant algebra of $\Lambda^\bullet \mathfrak{g}$ [31, 32, 35]. Here we study this cohomology at each fermion number beyond the cubic case. For $p = 3$, Chen’s Hamiltonian is a constant minus the quadratic Casimir [3]. Kostant’s results describe the relevant Casimir spectrum on the exterior algebra [4]. For odd $p \geq 5$ with $p \leq 2N - 1$, the Hamiltonian need not be determined by the quadratic Casimir, and the cohomology can be computed by linear algebra. The highest nonzero generator of the form $\text{Tr}(A^p)$ occurs at $p = 2N - 1$.

Generators and the condition $p \leq 2N - 1$. Let $V = \mathbb{C}^N$, and let $X = (X_{ij})$ be the generic anticommuting $N \times N$ matrix whose entries generate $\Lambda(V \otimes V^*)$. With $GL(V)$ acting by conjugation, the invariant subalgebra $\Lambda(V \otimes V^*)^{GL(V)}$ is generated by

$$\text{Tr} X, \quad \text{Tr} X^3, \quad \dots, \quad \text{Tr} X^{2N-1},$$

with no relations beyond anticommutativity [35]. For the traceless matrix A defined in (2), the generator $\text{Tr} X$ of degree one is absent. The anticommuting Cayley–Hamilton identity of Ref. [35] then implies $\text{Tr}(A^p) = 0$ for odd $p > 2N - 1$. Indeed, multiplying that identity by A^{2r} , taking the trace, and using the vanishing of even traces gives

$$N \text{Tr}(A^{2N-1+2r}) = 0, \quad r \geq 1.$$

This algebraic result has a cohomological counterpart. The Chevalley–Eilenberg theorem [31] gives

$$H^*(\mathfrak{su}(N); \mathbb{C}) \cong \Lambda(c_3, c_5, \dots, c_{2N-1}),$$

where c_{2m+1} , for $m = 1, \dots, N-1$, is the primitive cohomology class associated with the exponent m of $\mathfrak{su}(N)$ [31, 32]. Using the trace form to identify $\mathfrak{sl}_N$ with its dual, the class c_{2m+1} is represented, up to normalization, by $\text{Tr}(A^{2m+1})$ in the invariant exterior algebra [2, 35]. Under the standard transgression, the same class corresponds to the invariant polynomial $Y \mapsto \text{Tr}(Y^{m+1})$, where $Y \in \mathfrak{sl}_N$ is an ordinary commuting matrix variable [32]. Thus the cohomological description gives the nonzero traces in degrees $3, 5, \dots, 2N-1$, while the anticommuting Cayley–Hamilton identity gives their vanishing above this range. Therefore, for odd $p \geq 3$, $\text{Tr}(A^p)$ is nonzero exactly when

$$p \leq 2N-1. \quad (80)$$

Appendix B Data for each fermion number

The spectral computations at $N = 3$ and $N = 4$ used floating-point arithmetic with double precision and a decomposition by singular values of the stacked matrix A_R in (39). Fock states are ordered lexicographically by bit pattern, and the Q_p matrices are built by explicit enumeration of all closed index chains in $\text{Tr}(\Psi^p)$, equivalently $\text{Tr}(A^p)$ by Theorem 2. Ranks were determined with a threshold of 10^{-10} relative to the largest singular value. At $N = 3, 4$, the zero and nonzero clusters are separated by more than four orders of magnitude.

The $N = 5$ computations used sparse matrix assembly, with inner loops accelerated by Numba, to build each sector map Q_R as a compressed sparse row matrix. The Gram matrix $Q_R^\top Q_R$ was then decomposed into connected components via its sparsity graph. Each block was diagonalized independently using dense `eigvalsh` or $\text{LDL}^\top$ factorization. In every block, the zero and nonzero eigenvalue clusters are separated by more than two orders of magnitude. A second implementation reproduced all reported $N = 5$ ranks. It finds the connected components by a disjoint set algorithm and counts the $\text{LDL}^\top$ pivots in each block.

The complete rank arrays for the four cases computed in every sector are supplied in the ancillary file `rank_data.json`. For $(5, 3)$, $(5, 4)$, and $(7, 4)$, `verify_ranks.py` reconstructs the sector maps and performs the checks over finite fields described in Section 4.1. For $(5, 5)$, the entries through $R = 10$ record the numerical block computations described above. The remaining nontrivial ranks follow from the symmetry in Eq. (15). The script does not attempt the $N = 5$ case.

B.1 Energy levels: quintic model, $p = 5$, $N = 4$

The table gives the spectrum through half filling, $0 \leq R \leq 8$. Blank cells denote zero multiplicity.

R	$E = 0$	225	600	900	1100	1400	2100	2400	3600	dim
0	0								1	1
1	0							15	1	16
2	0					90		30		120
3	0		245		84	195	20	15	1	560
4	125	512	574	175	258	120	55		1	1820
5	1375	1536	497	525	354	15	65		1	4368
6	5000	1536	497	525	354	15	65	15	1	8008
7	9625	512	574	175	258	210	55	30	1	11440
8	11750		490		168	390	40	30	2	12870

Sectors $R = 9, \dots, 16$ follow by particle–hole conjugation. The total BPS count is 44,000.

B.2 Other computed energy levels

Quintic model at $N = 3$. The distinct energies are $\{0, 225, 600\}$, and the BPS multiplicities are given in (41).

Model of degree seven at $N = 4$. The distinct energies are

$$\{0, 1764, 4410, 7350, 7840, 14700, 18816, 35280\}.$$

The BPS multiplicities are given in (48). The total BPS count is 59,136.

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